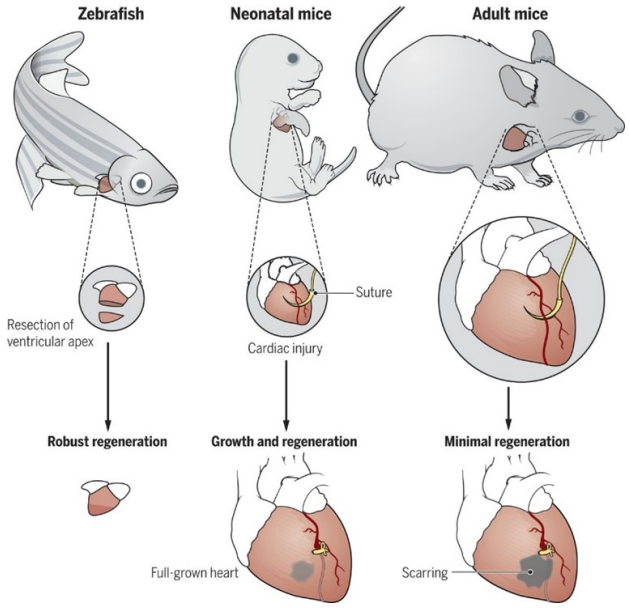
Decoding Postnatal Heart Maturation

From a Spatial Atlas to In Vivo Functional Genomics

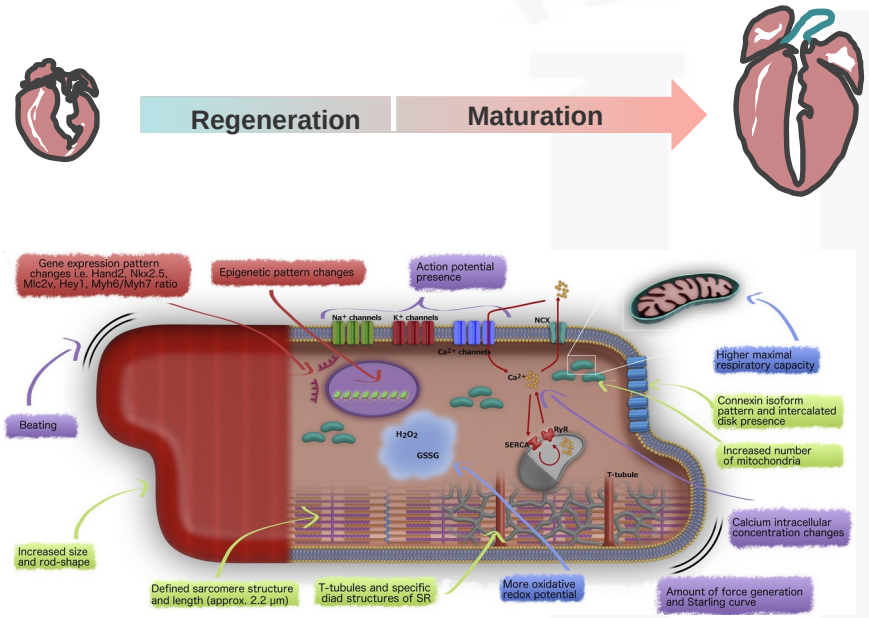
Jiandong Liu, Ph.D.
McAllister Heart Institute
Department of Pathology and Laboratory Medicine
University of North Carolina at Chapel Hill

Wang, Dong, Song et al.
Nature Cardiovascular Research, 2026

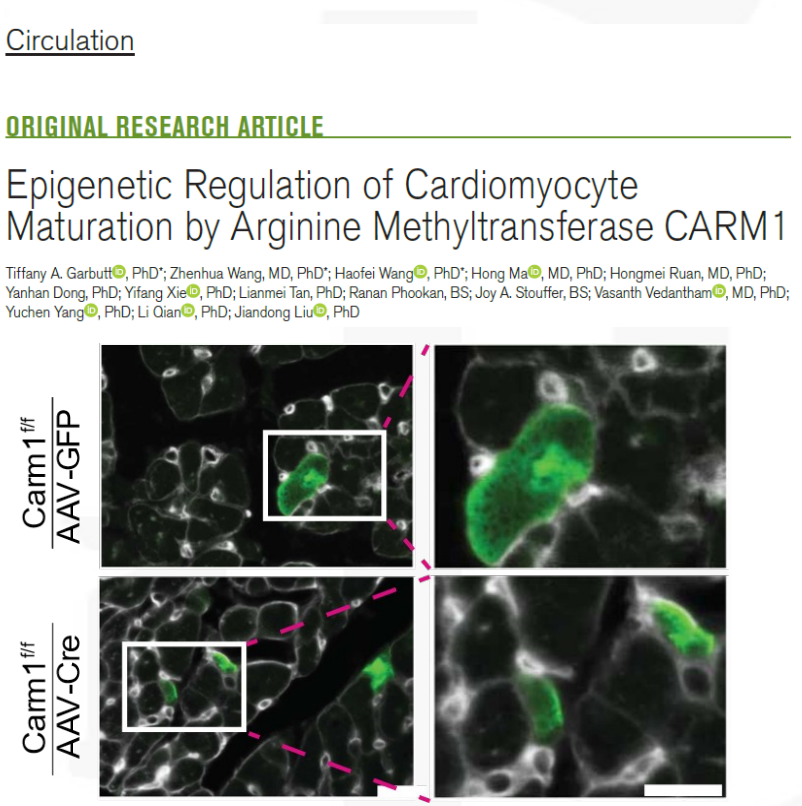
Cardiomyocyte maturation is a coordinated functional transition



Tzahor E, Poss KD. Science. 356:1035-1039

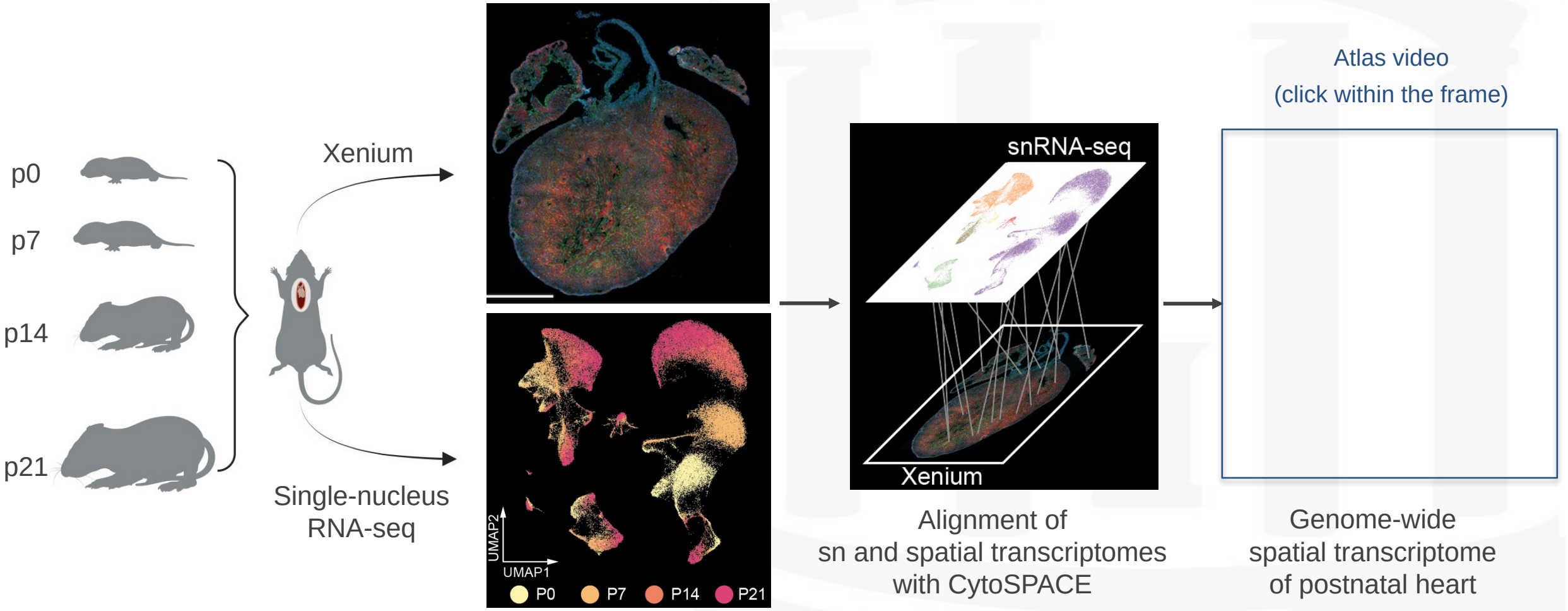


Kolanowski T et al., Int. J. Cardiol. 241:379-386
Guo Y and Pu WT. Circ Res. 126:1086-1106
Doyle MJ et al., Stem Cell Rev Rep.11:710-27.

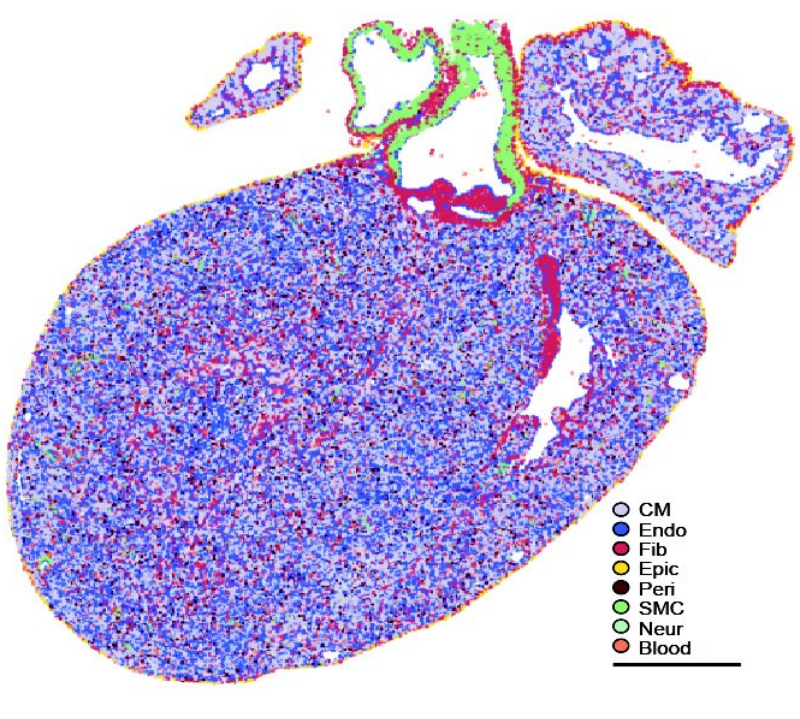


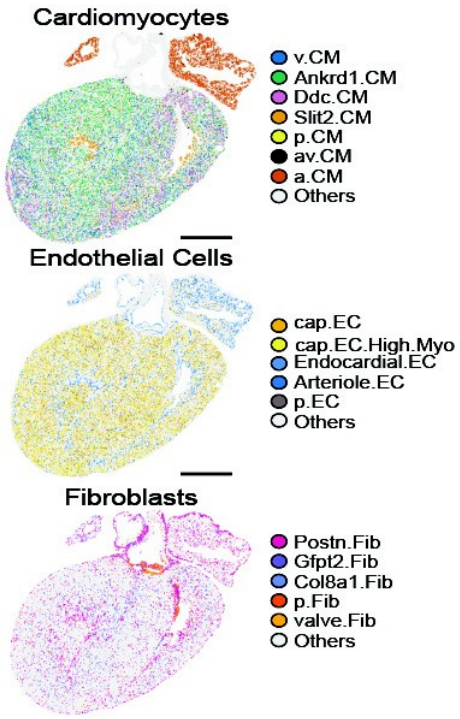
Garbutt T et al., Circulation. 2024

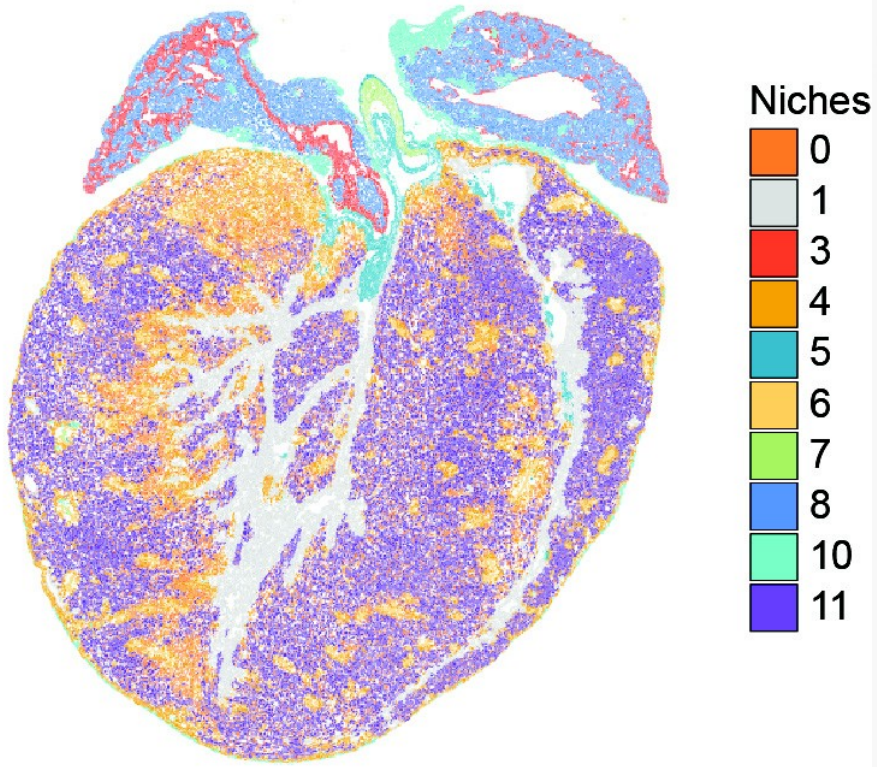
An integrated atlas captures the developing postnatal heart



Major cardiac lineages form organized spatial neighborhoods



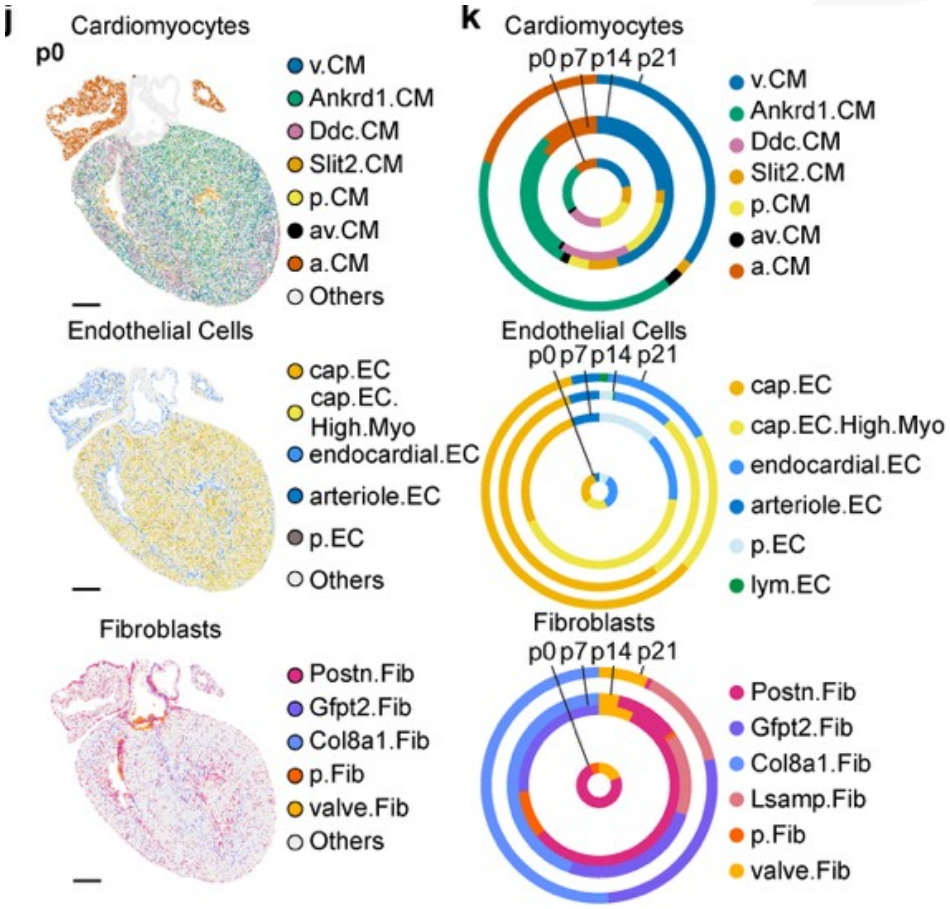




Cell identity and tissue position provide complementary biological information.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 1f,j; Fig. 2a

Cell states have distinct locations and developmental dynamics



CM states

Regional specialization

EC states

Vascular identity

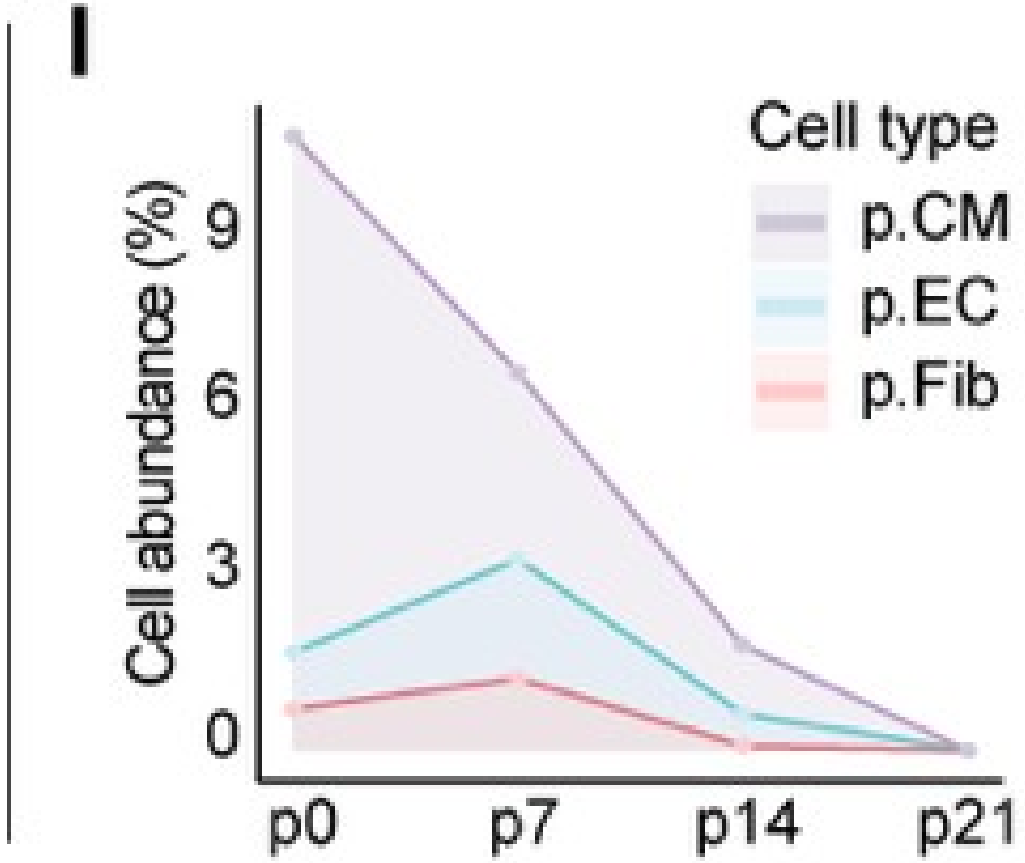
Fibroblast states

Matrix and anatomical programs

The atlas resolves 25 states with distinct transcriptional and spatial identities.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 1j,k

Proliferative states follow different postnatal schedules



Early CM cell-cycle activity

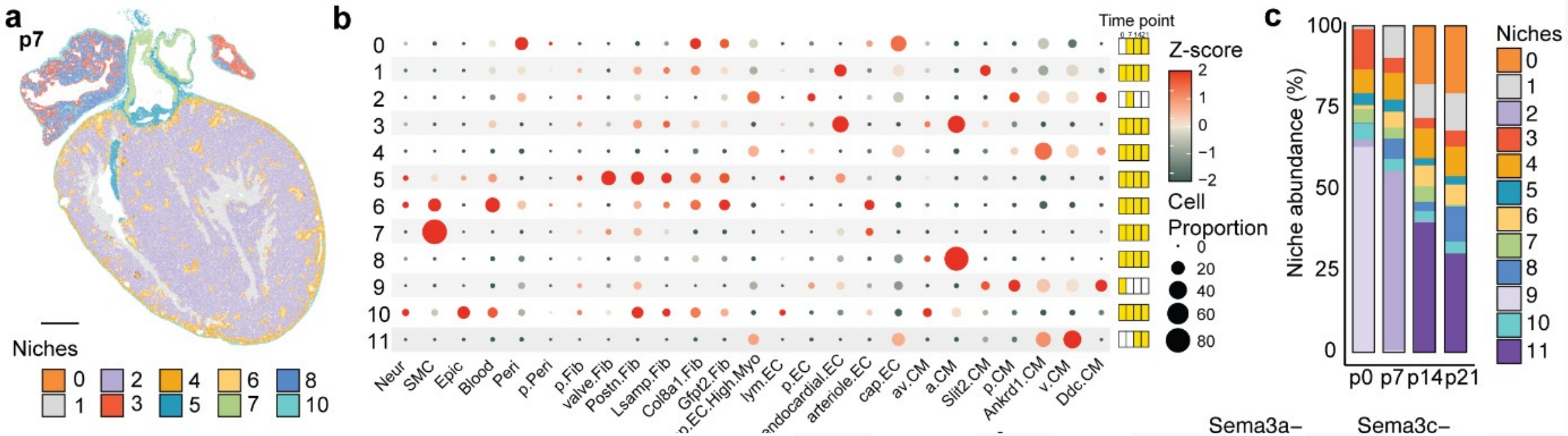
Transient fibroblast proliferation

Stage-dependent vascular growth

CMs, endothelial cells and fibroblasts have distinct developmental growth dynamics.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 1l; Supplementary Fig. 10b

Twelve cellular niches capture regional heart development



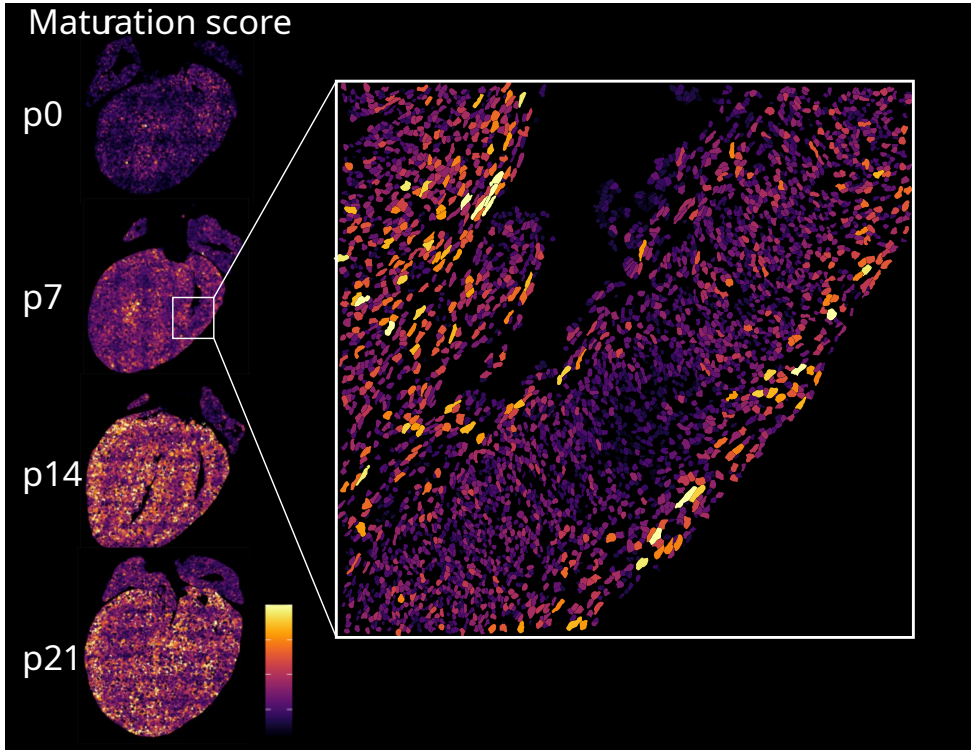
8 persistent niches and 4 ventricular stage-specific niches

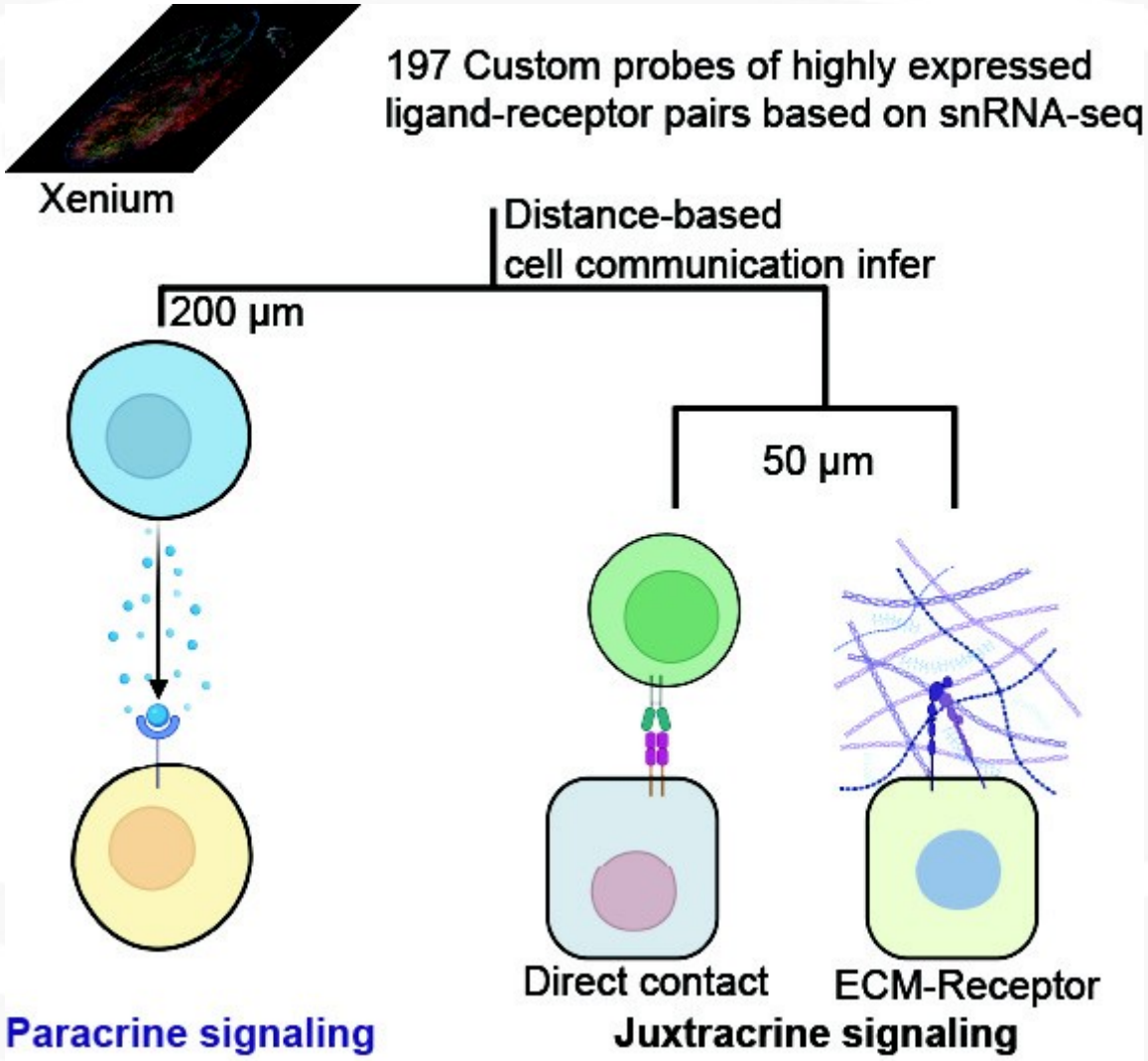
A capillary/pericyte-rich niche emerges at P7 and expands

Development remodels local cellular combinations as well as individual cell states.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 2a–c

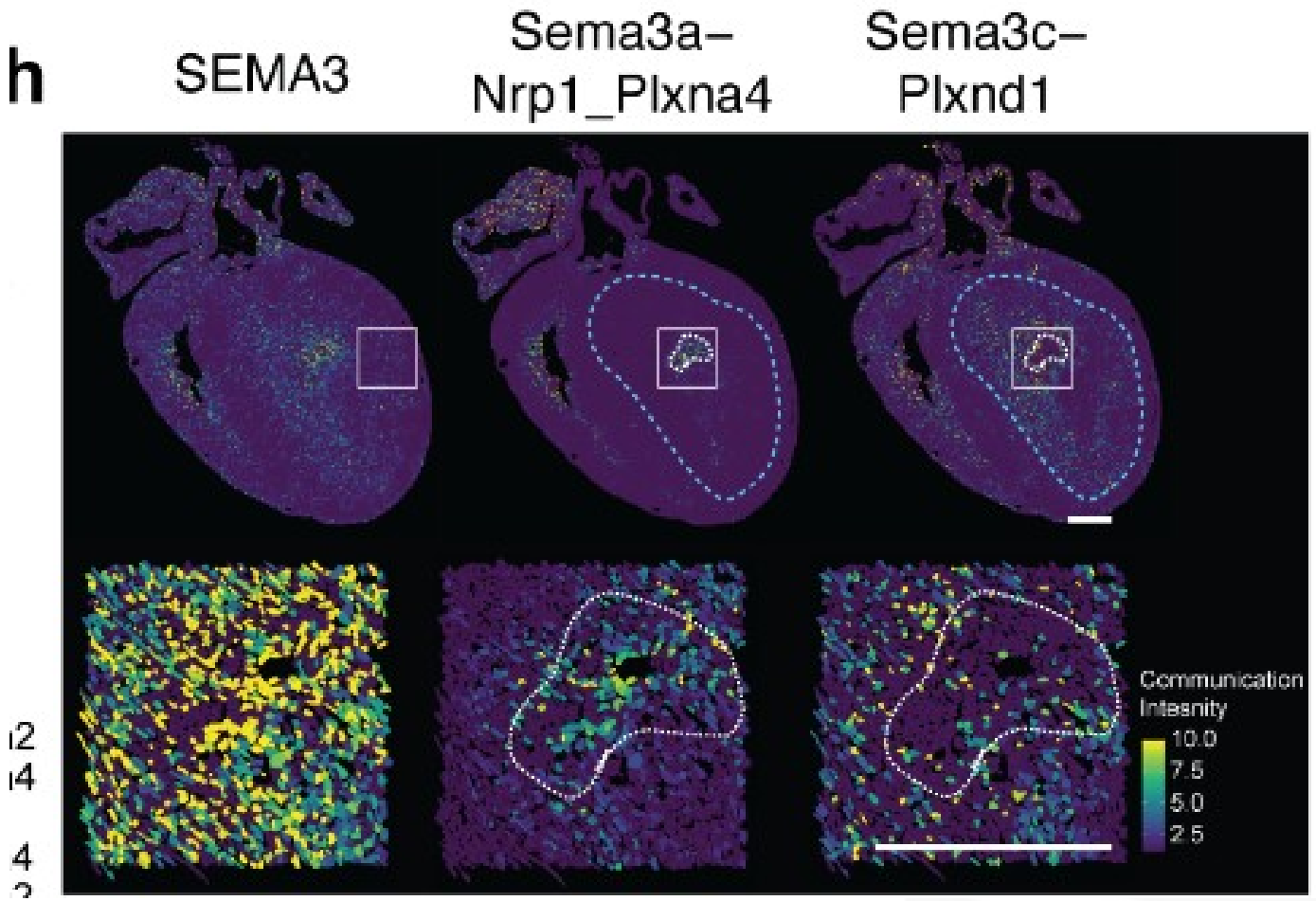
Spatial constraints refine cell–cell communication inference





Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 2d; original communication schematic

Related signaling pairs can occupy different cardiac territories

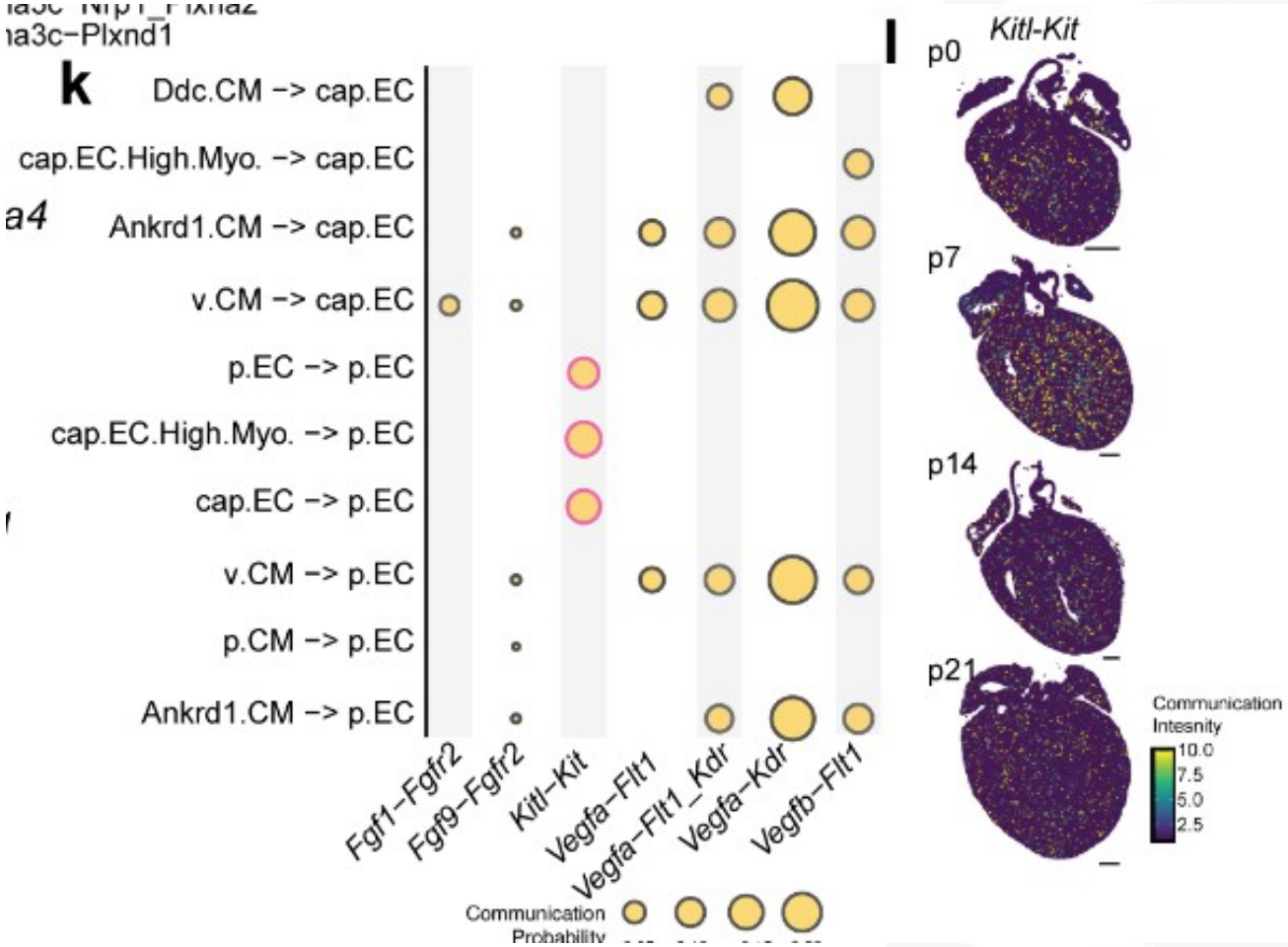


- SEMA3 signaling
- Different ligand–receptor pairs
- Different ventricular territories

Spatially resolved interactions reveal local specificity within a signaling family.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 2g,h

A localized vascular cue highlights the proliferative EC state



KITL–KIT

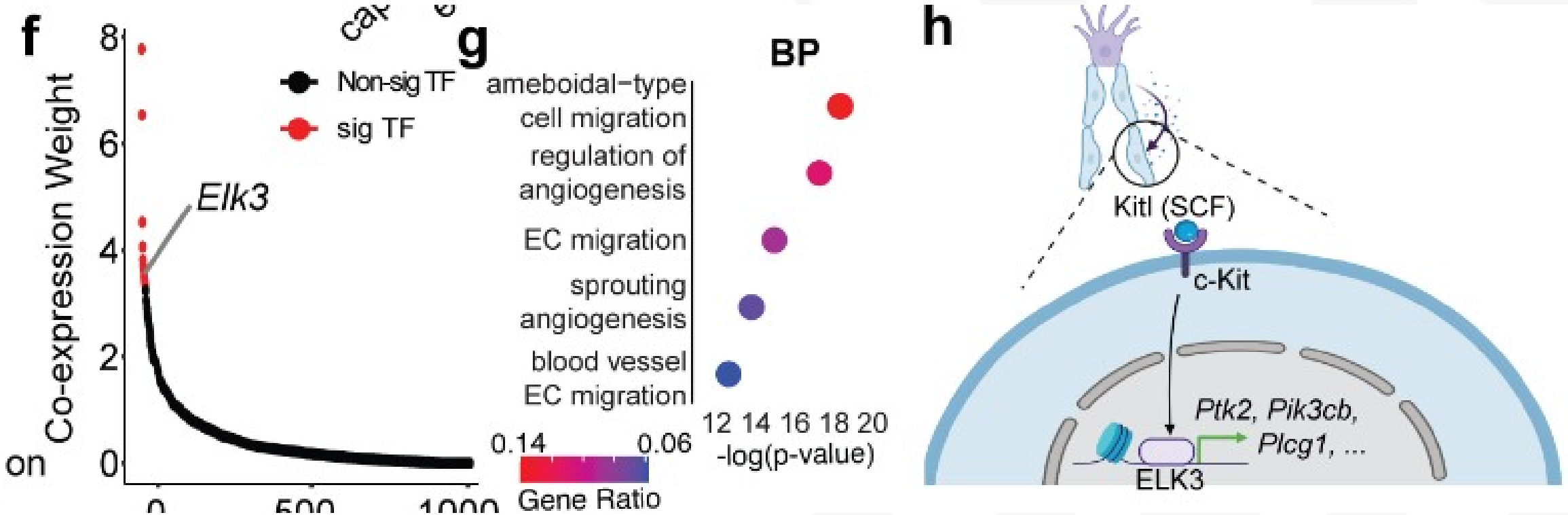
Proliferative EC state

Peak inferred activity at P7

KITL–KIT is a candidate cue associated with the P7 proliferative EC state.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 2k,l

SigNET links candidate extracellular cues to regulatory programs



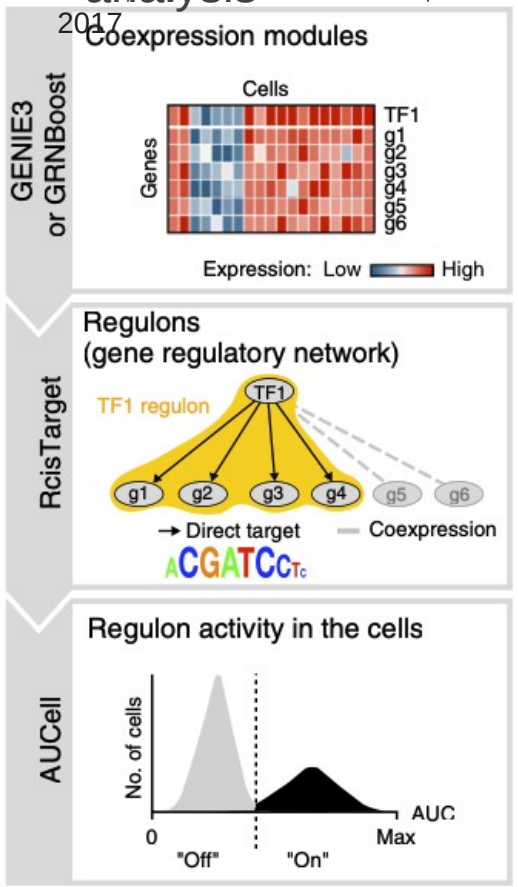
Spatial signal → receptor-associated response → candidate TF program

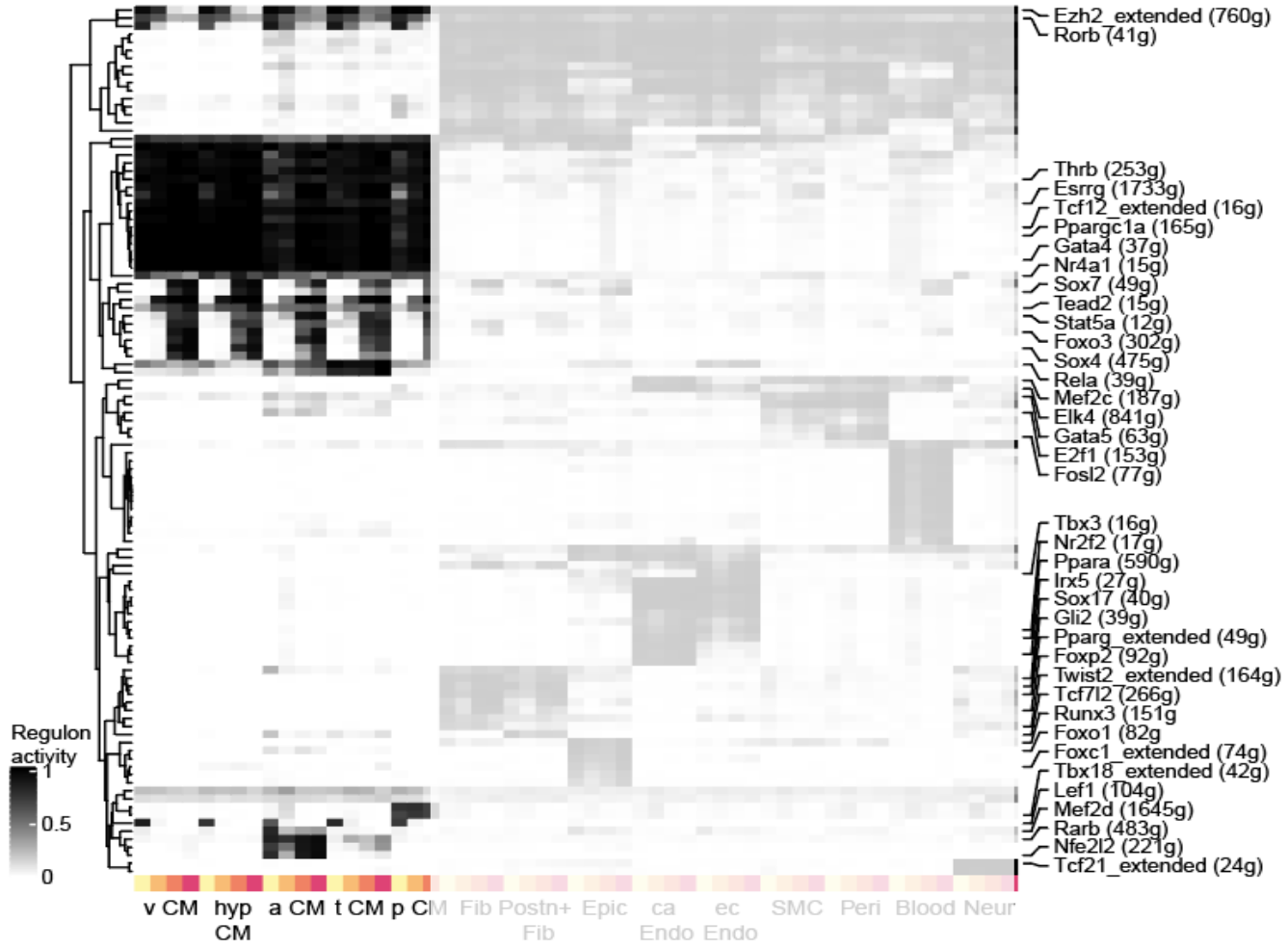
A signaling-to-transcription framework makes spatial observations experimentally actionable.

CM regulatory programs change sharply between P7 and P14

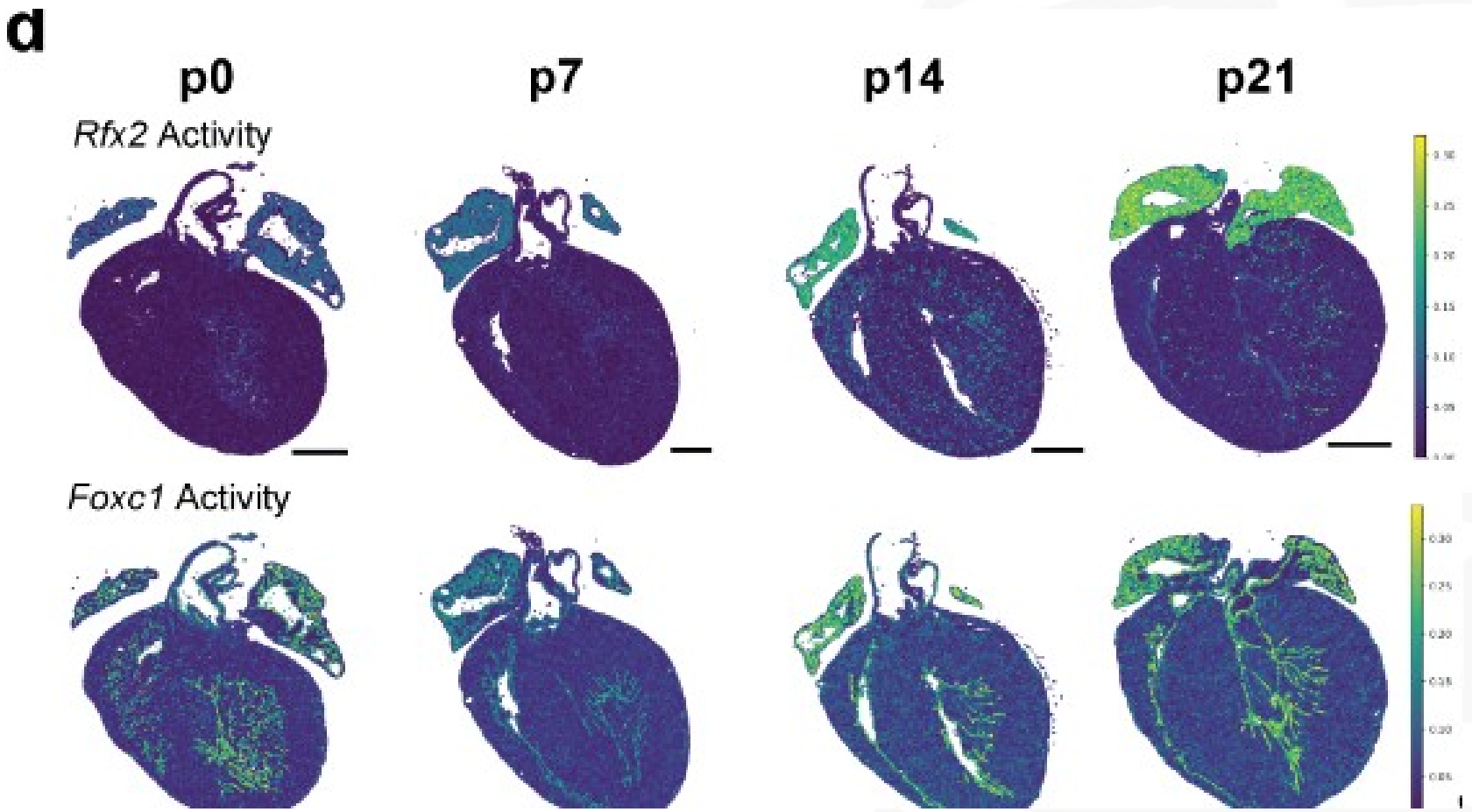
SCENIC analysis

Aibar, S. et al. *Nat Methods*, 2017





Regulatory programs also retain regional identity

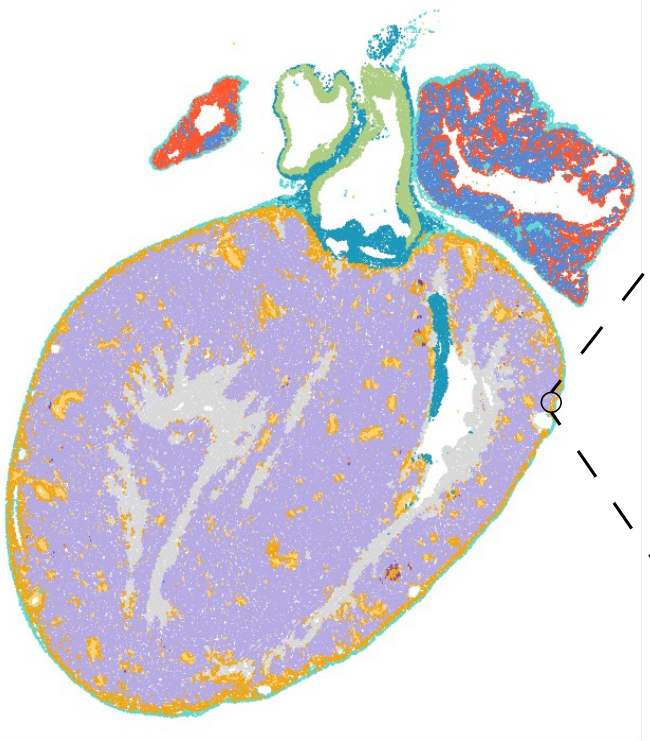


Rfx2
Atrial activity

Foxc1
Inner lining / outflow tract

Time and anatomical position jointly organize the regulatory landscape.

CM maturation varies across both age and tissue position



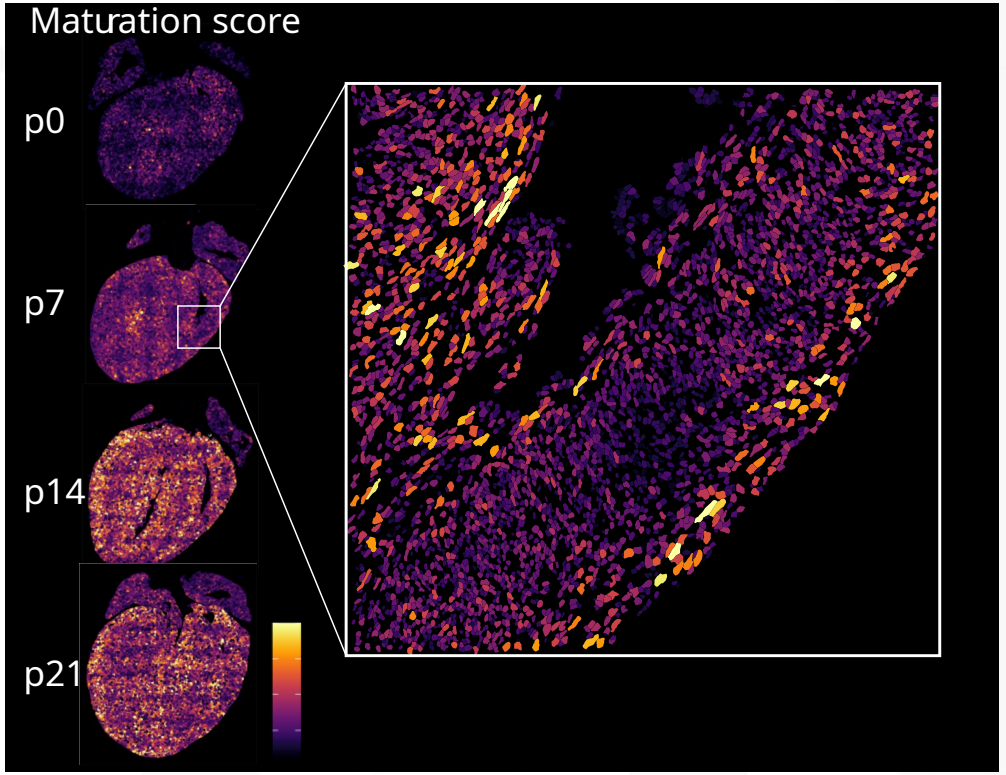
Maturation-related Gene

TFs
Rxrg
Pargc1a
Pparcg1b
Cebpa
Cebpb
Esrra
Esrrb
Esrrg
Mitf

Electrophsiology
Pln
Kcnq1
Jph1
Jph2
Bin1
Lpin1

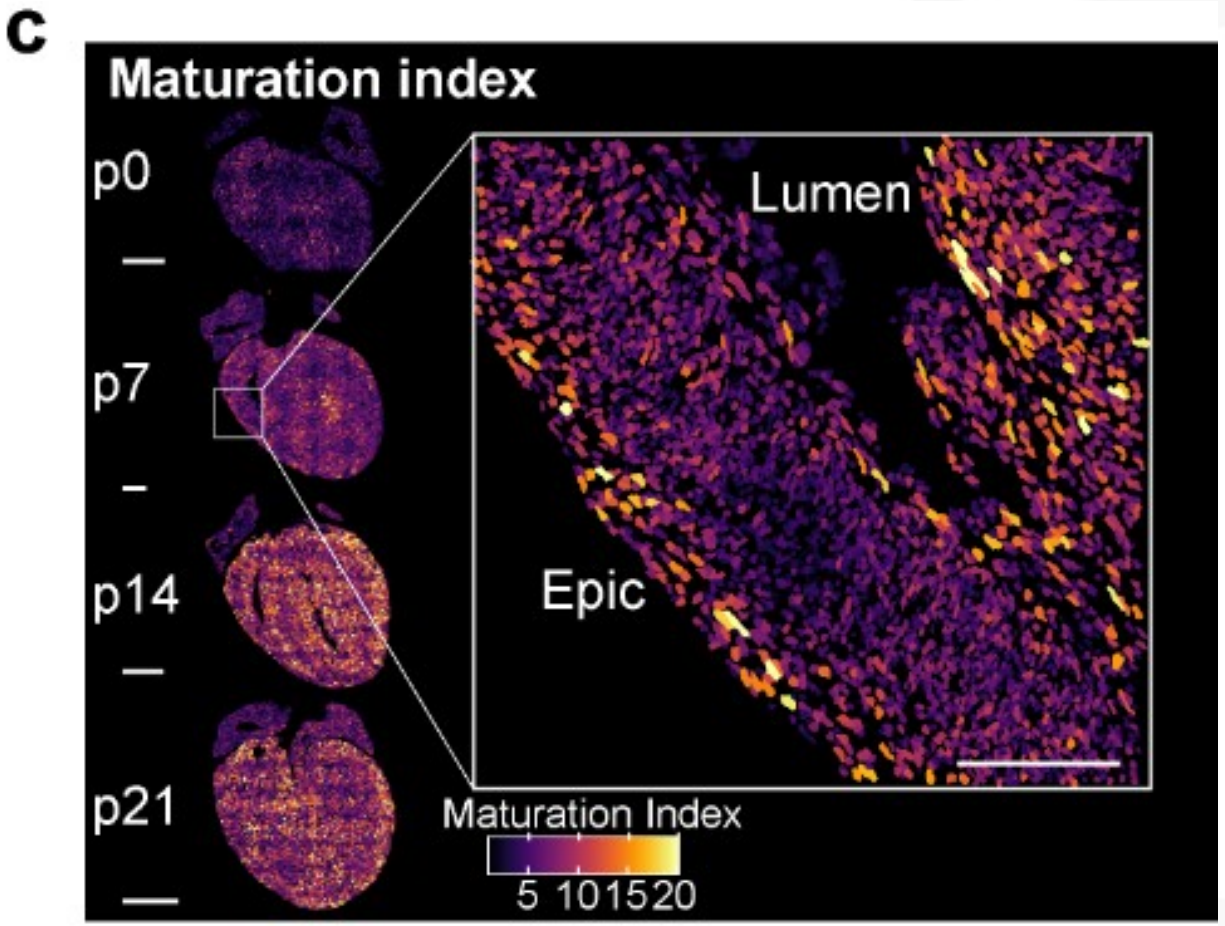
Sacromere
Myh6
Tnni3
Myl2

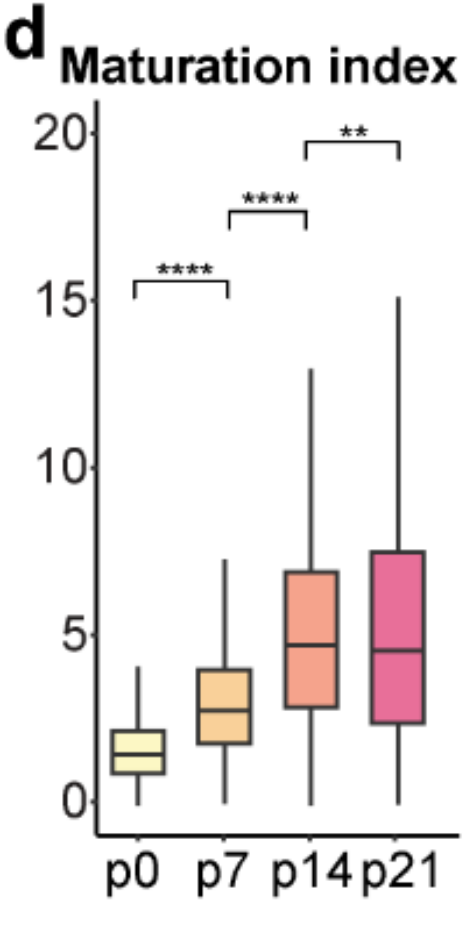
Metabolism
Ppara
Ppard
Cpt1b
Pargc1a



21 measured Xenium maturation genes · Seurat AddModuleScore (published Methods)

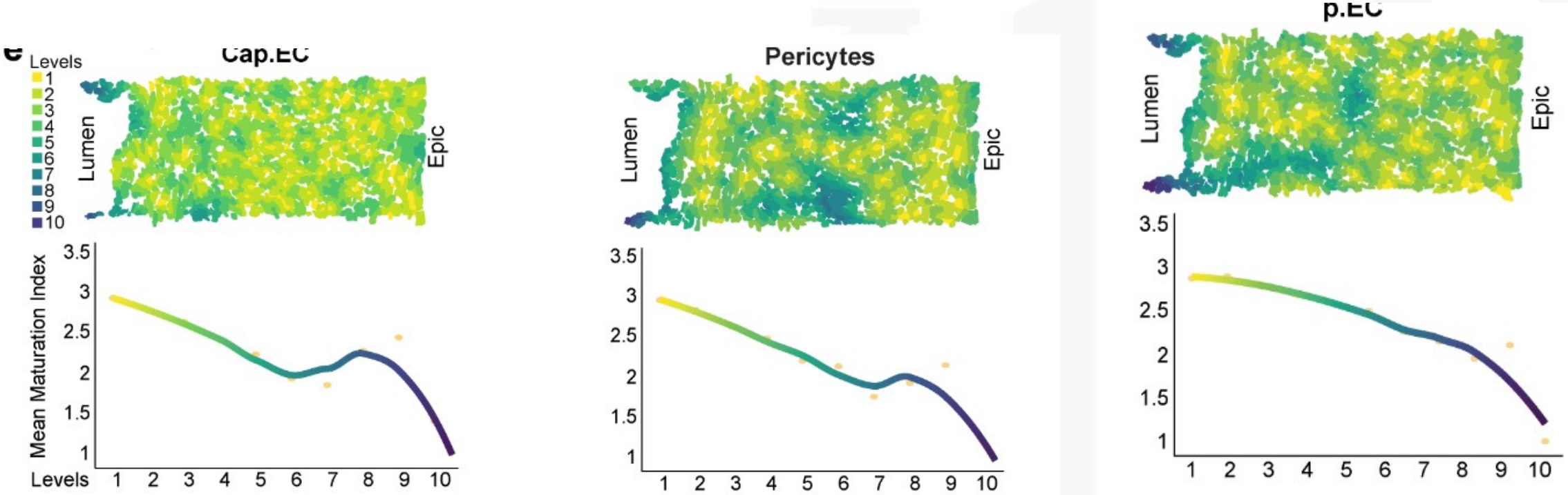
The measured maturation index increases through postnatal development





Age-related progression coexists with substantial within-section heterogeneity.

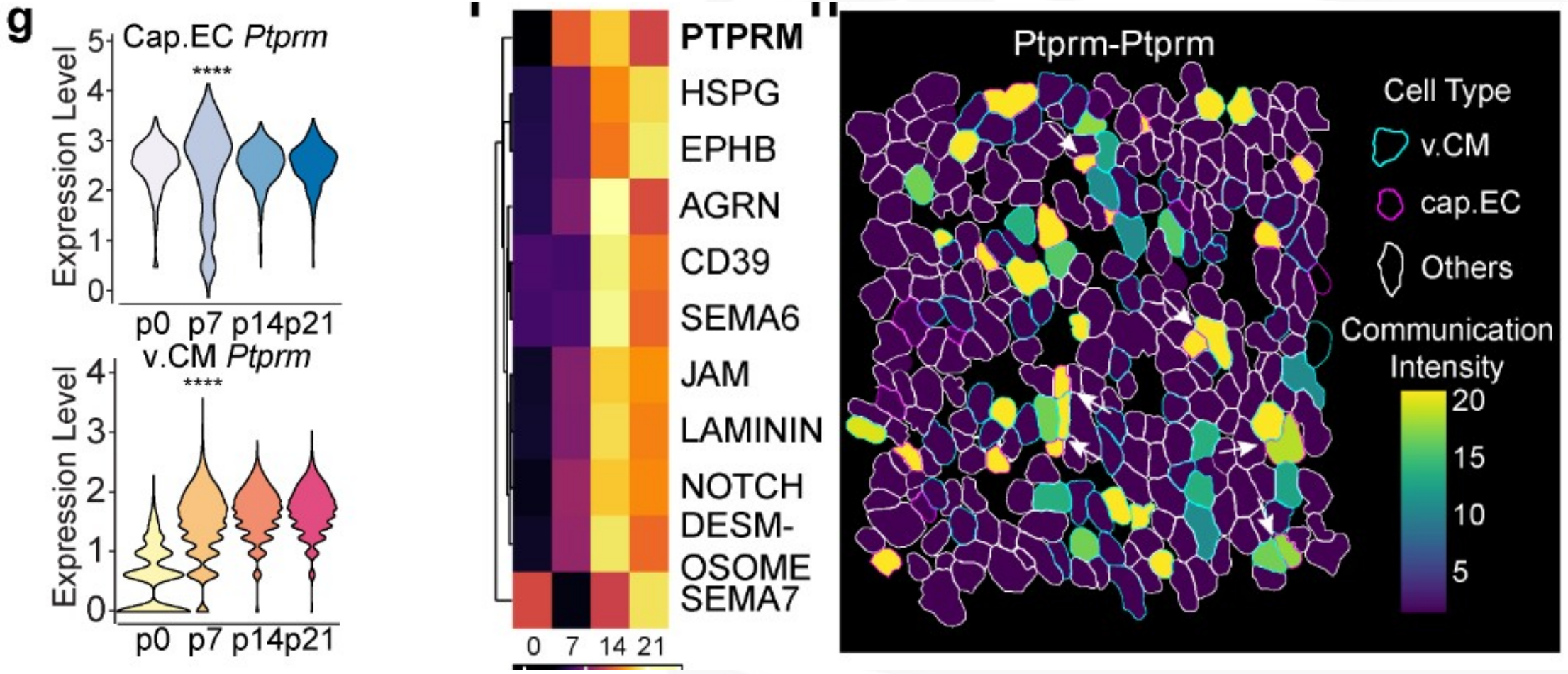
Higher CM maturation scores occur near the microvasculature



At P7, the microvascular neighborhood is associated with higher CM transcriptional maturation.

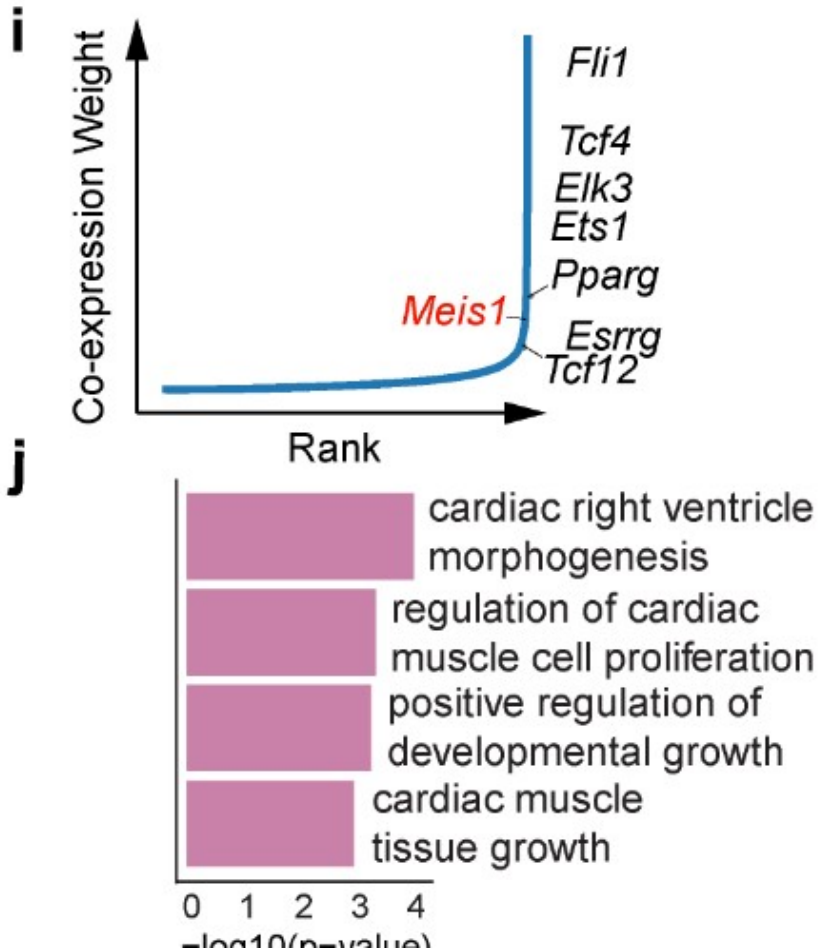
Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 4e

PTPRM emerges as a contact-associated maturation candidate



Expression, inferred interaction and boundary protein localization converge on PTPRM.

A PTPRM-associated regulatory link points to MEIS1



SigNET nominates MEIS1

CM cell-cycle-exit program

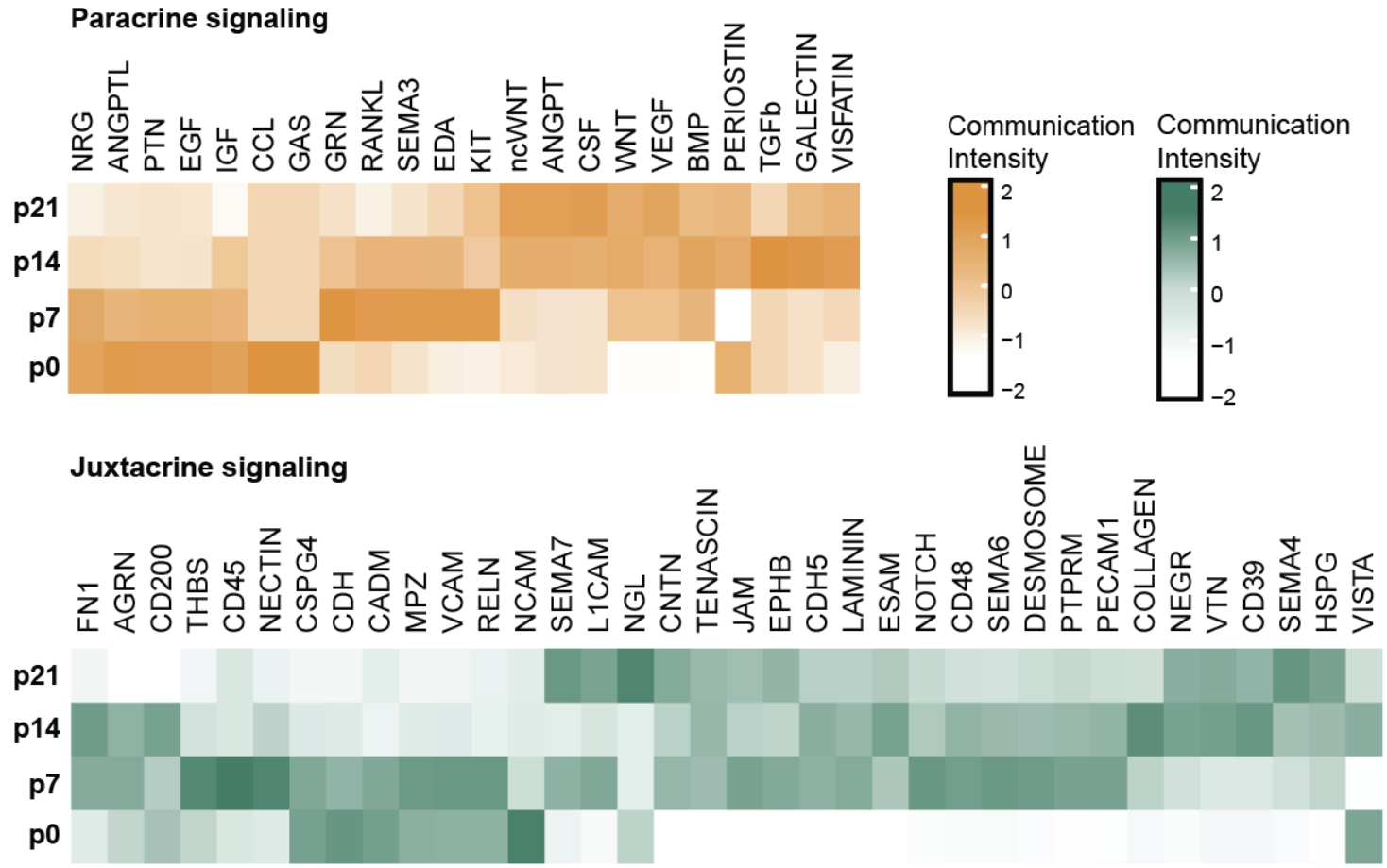
At P8, Ptprm perturbation:

↑ Cell-cycle-associated RNA

↓ Meis1 and maturation RNA

Targeted Ptprm perturbation supports a role in maturation-associated gene regulation.

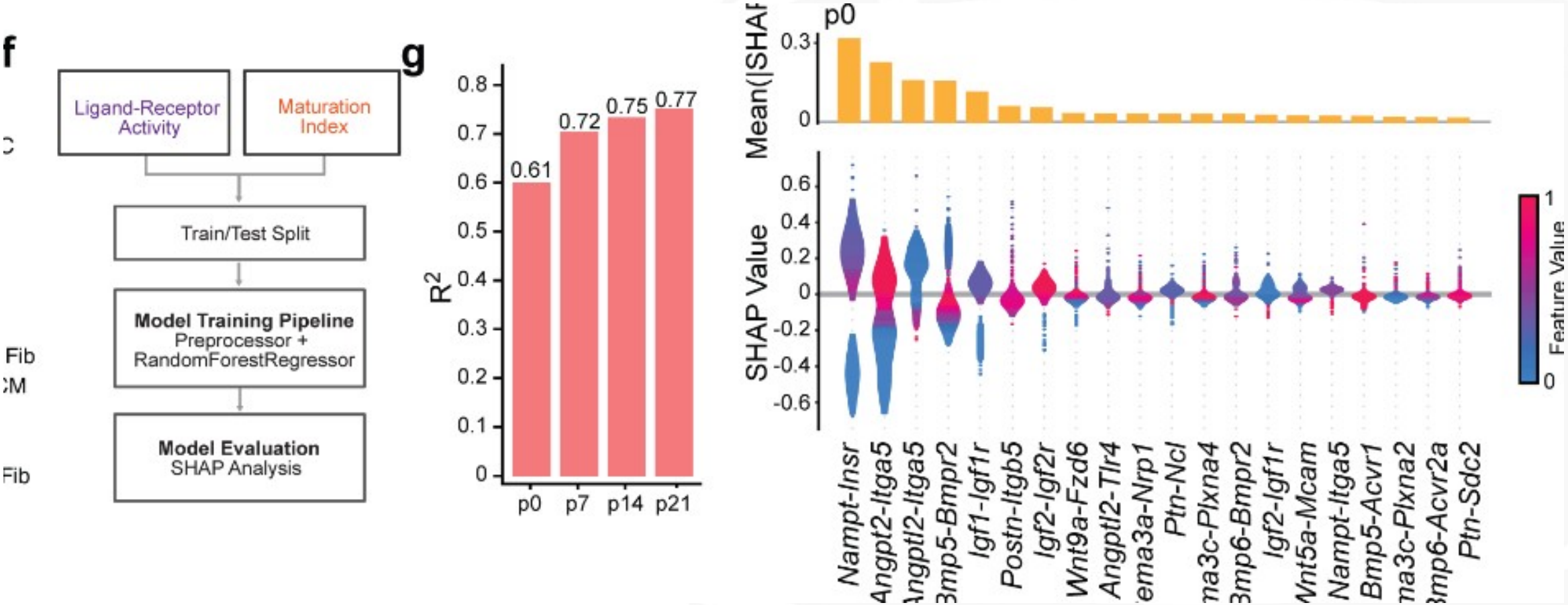
CMs encounter changing extrinsic cues during maturation



The extracellular signaling landscape changes together with CM state.

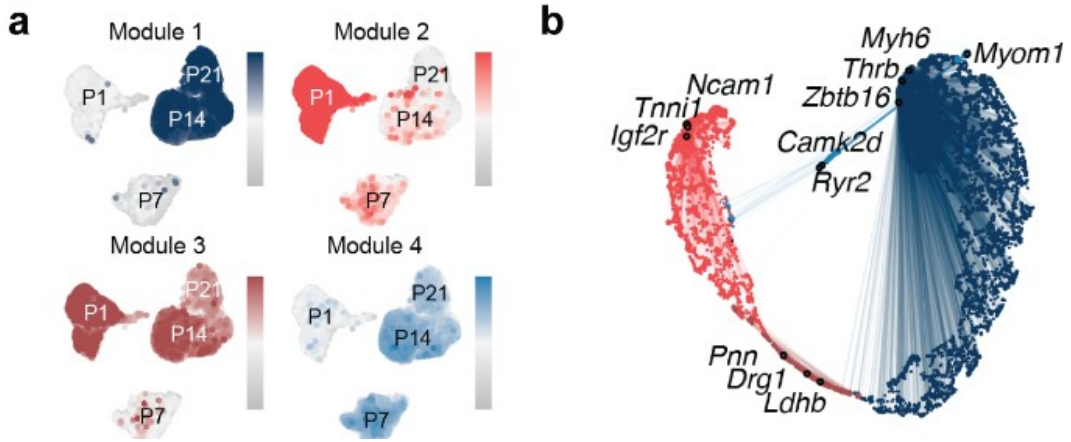
Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 5e; Supplementary Fig. 21; original signaling heatmaps

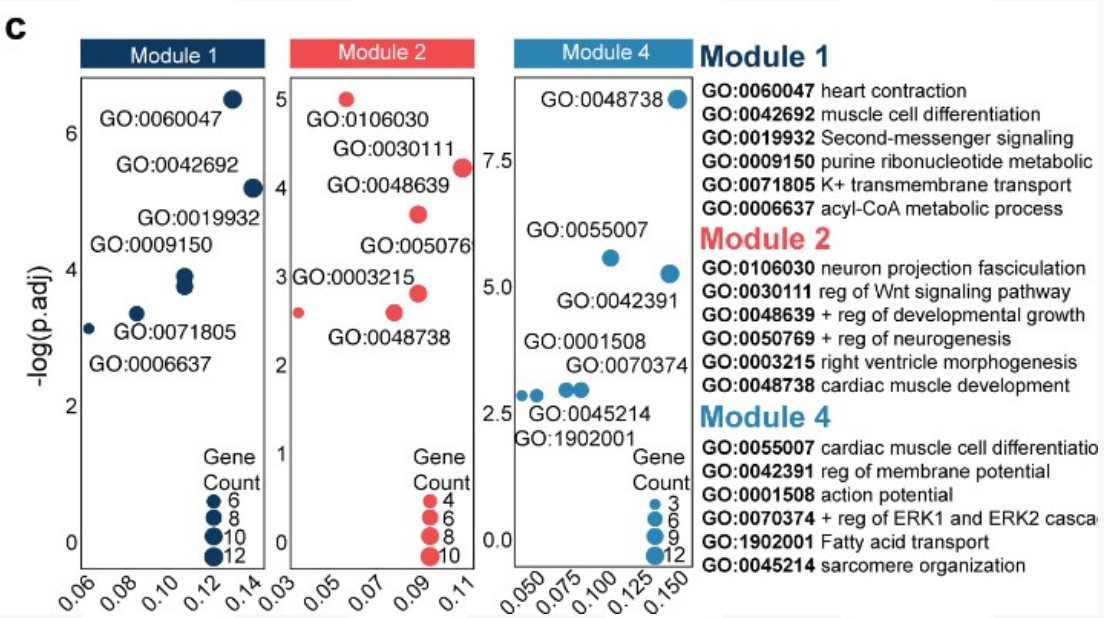
Signaling activities predict maturation scores and prioritize pathways



Prediction provides candidate prioritization for functional testing.

CM gene modules capture distinct aspects of maturation





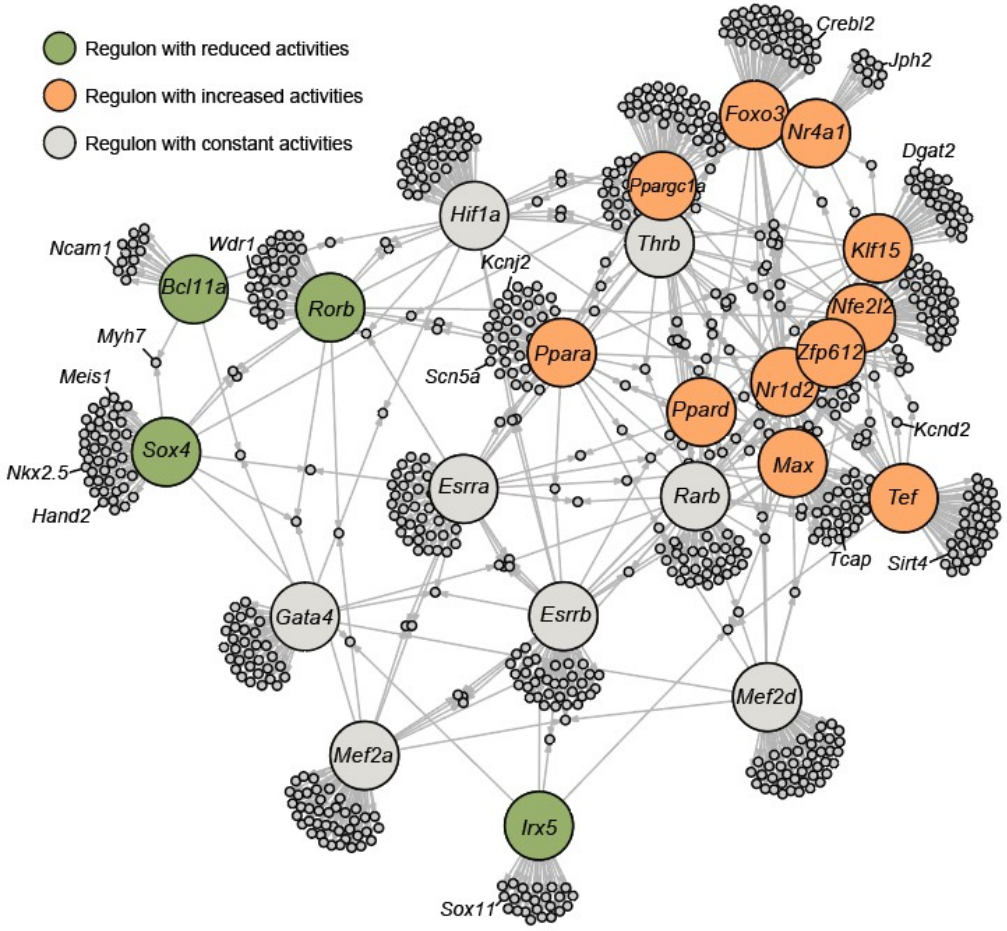
Contraction

Membrane-potential regulation

Fatty-acid transport

Maturation comprises multiple coordinated transcriptional programs.

Temporal regulon networks nominate intrinsic maturation regulators

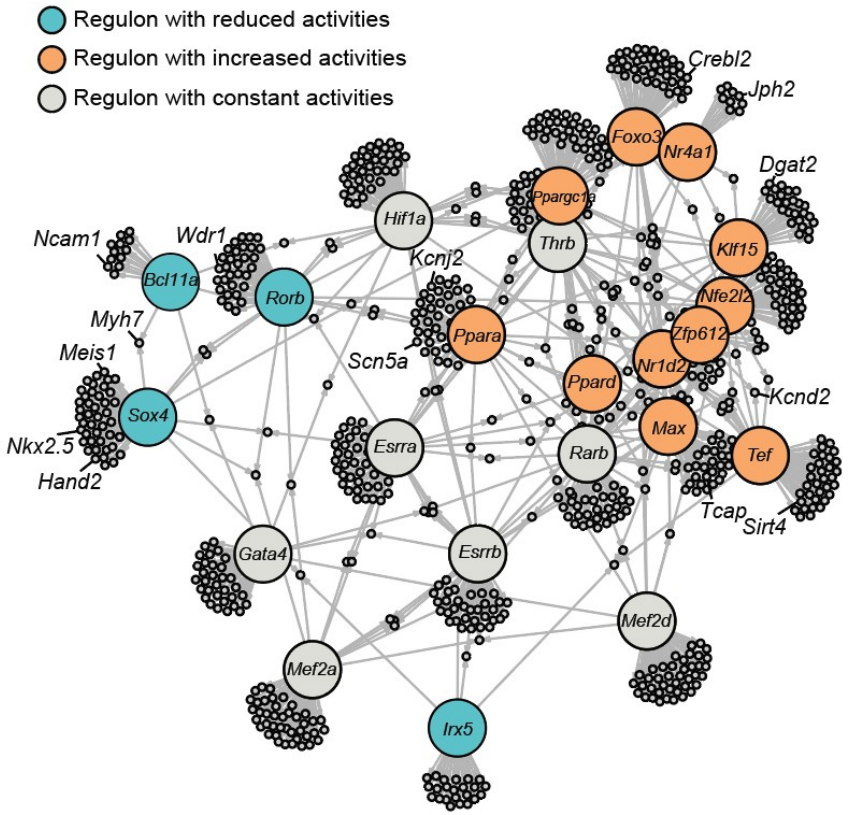


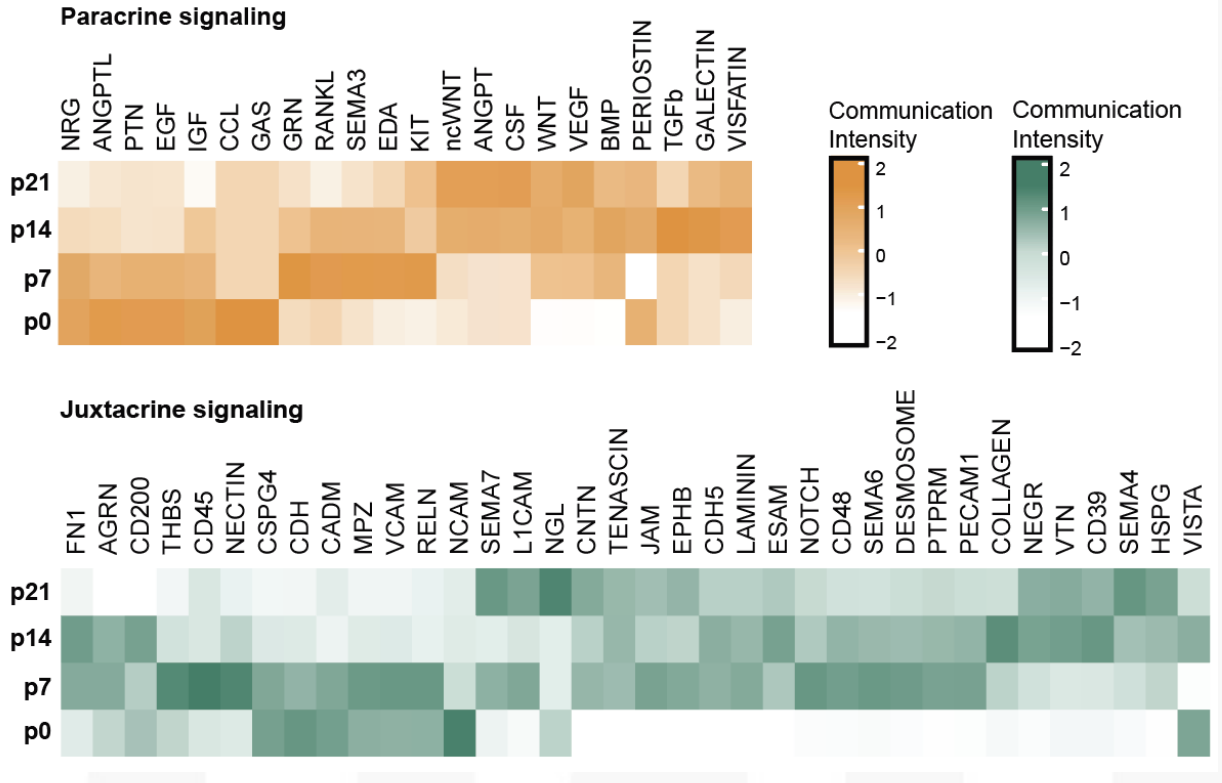
32 CM regulons

7 previously linked
25 candidates

32 CM-specific regulons include 7 previously linked and 25 candidate maturation regulators.

The atlas nominates regulators that now require in vivo tests





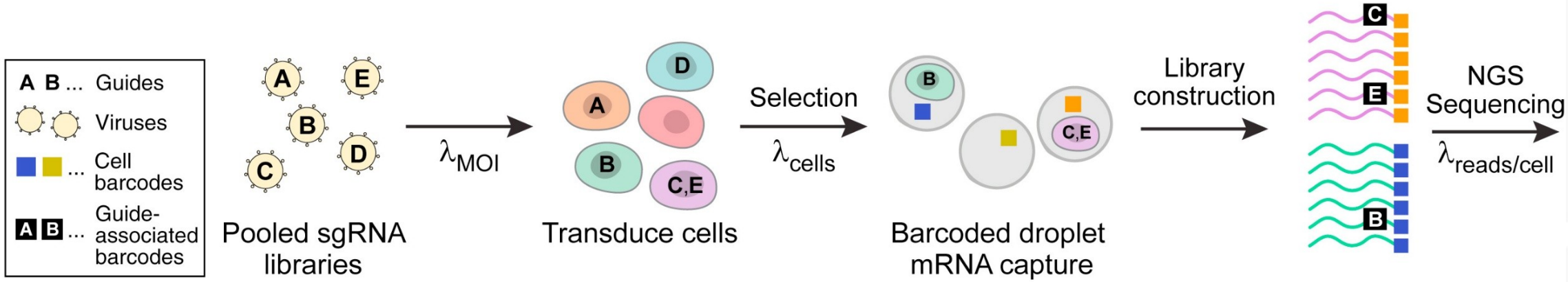
How can we test the functions of uncharacterized regulators efficiently and directly *in vivo*?

Pooled perturbation links gene identity to an RNA phenotype

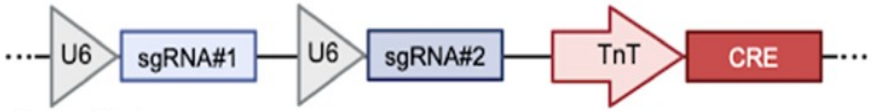
Perturb-Seq: Dissecting Molecular Circuits with Scalable Single-Cell RNA Profiling of Pooled Genetic Screens

Atray Dixit,[1,2,9] Oren Parnas,[1,9,10] Biyu Li,[1] Jenny Chen,[1,2] Charles P. Fulco,[1,4] Livnat Jerby-Arnon,[1] Nemanja D. Marjanovic,[1,3] Danielle Dionne,[1] Tyler Burks,[1] Raktima Raychowdhury,[1] Britt Adamson,[5] Thomas M. Norman,[5] Eric S. Lander,[1,4,6] Jonathan S. Weissman,[5,7] Nir Friedman,[1,8] and Aviv Regev[1,6,7,11,*]

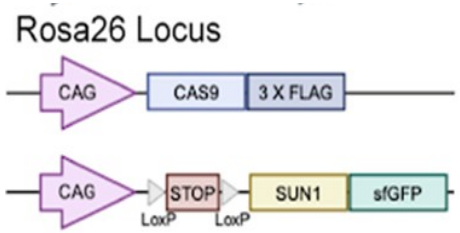
Application:
discovery of gene function
mapping transcriptional networks
dissection of genetic interaction



PIP-seq tests candidate regulators within the postnatal heart



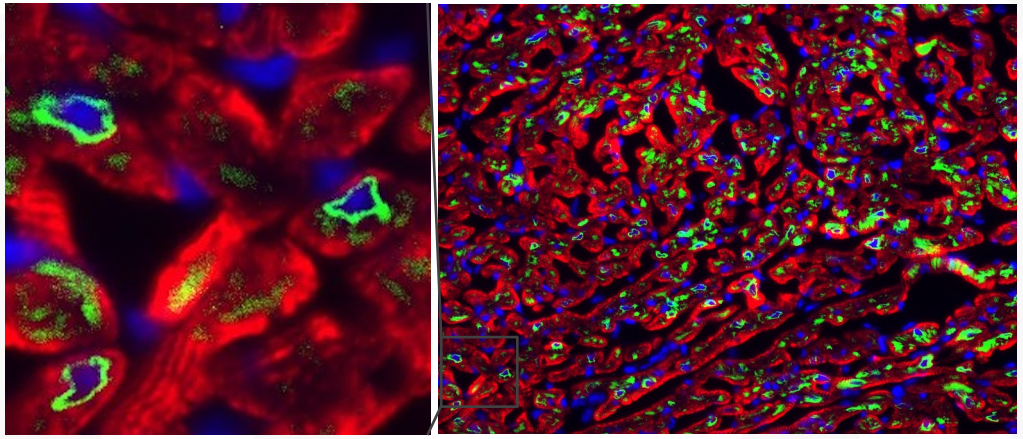
AAV9 with dual sgRNAs and CM-specific CRE



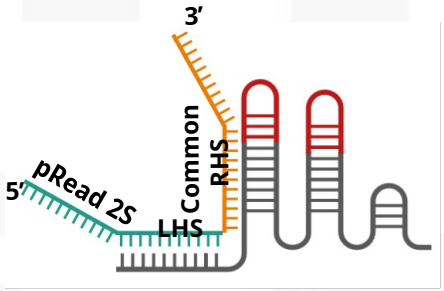


CM nuclei membrane labeling of AAV9-infected cells

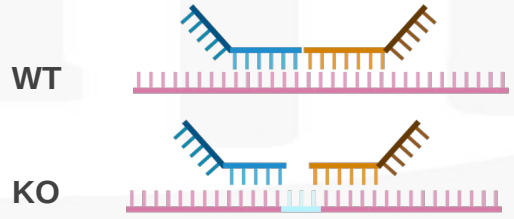
ACTN2/SUN1-GFP/DAPI



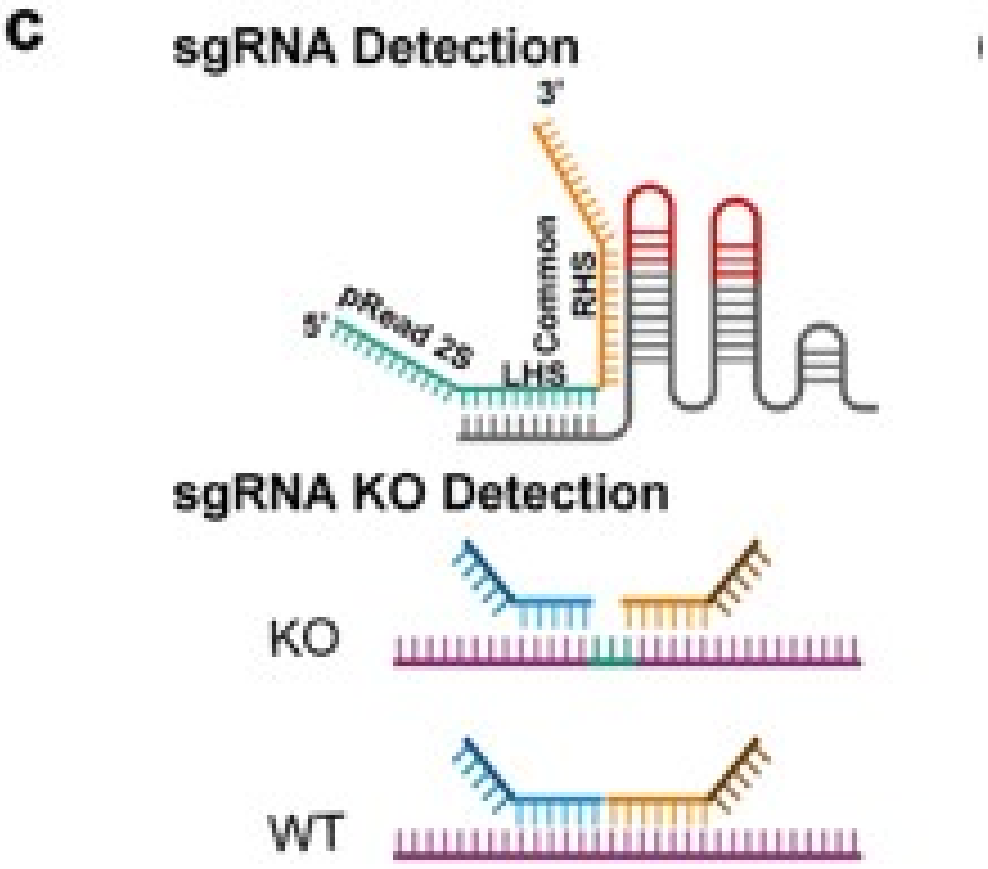
Direct capture of sgRNA with probe hybridization



Perturbation detection probe



Custom probes improve guide recovery and perturbation evidence



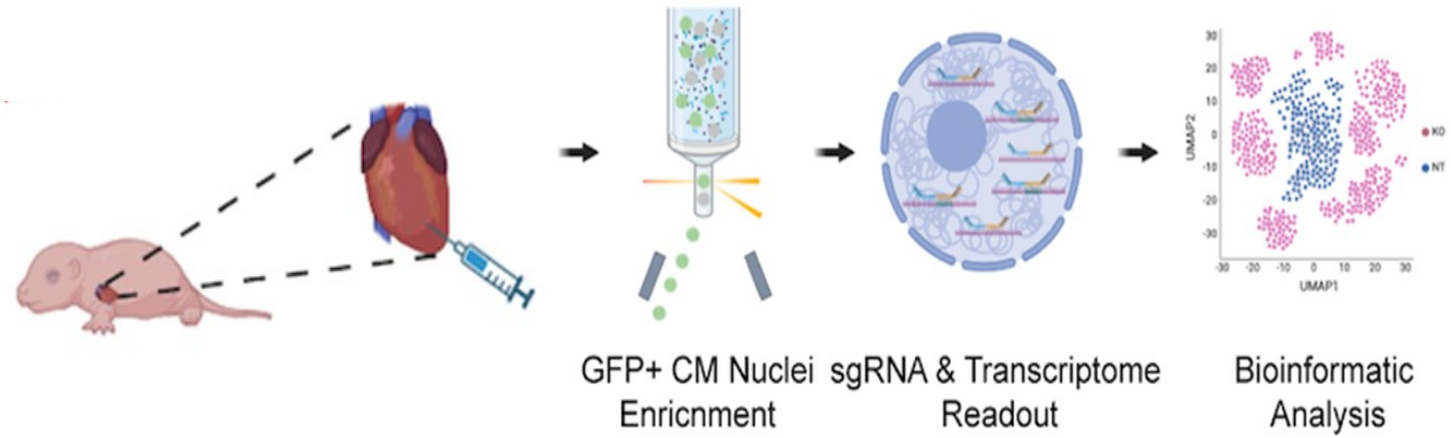
Guide detection
Identify the perturbation

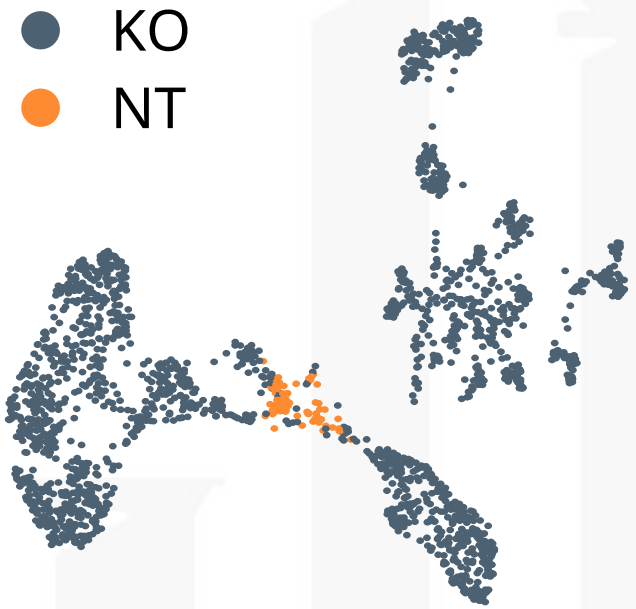
Target-site probe behavior
Additional evidence consistent with editing

Guide identity and target-probe behavior provide complementary information.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 6c; Supplementary Fig. 24d

A pooled neonatal screen captures thousands of perturbed CM nuclei

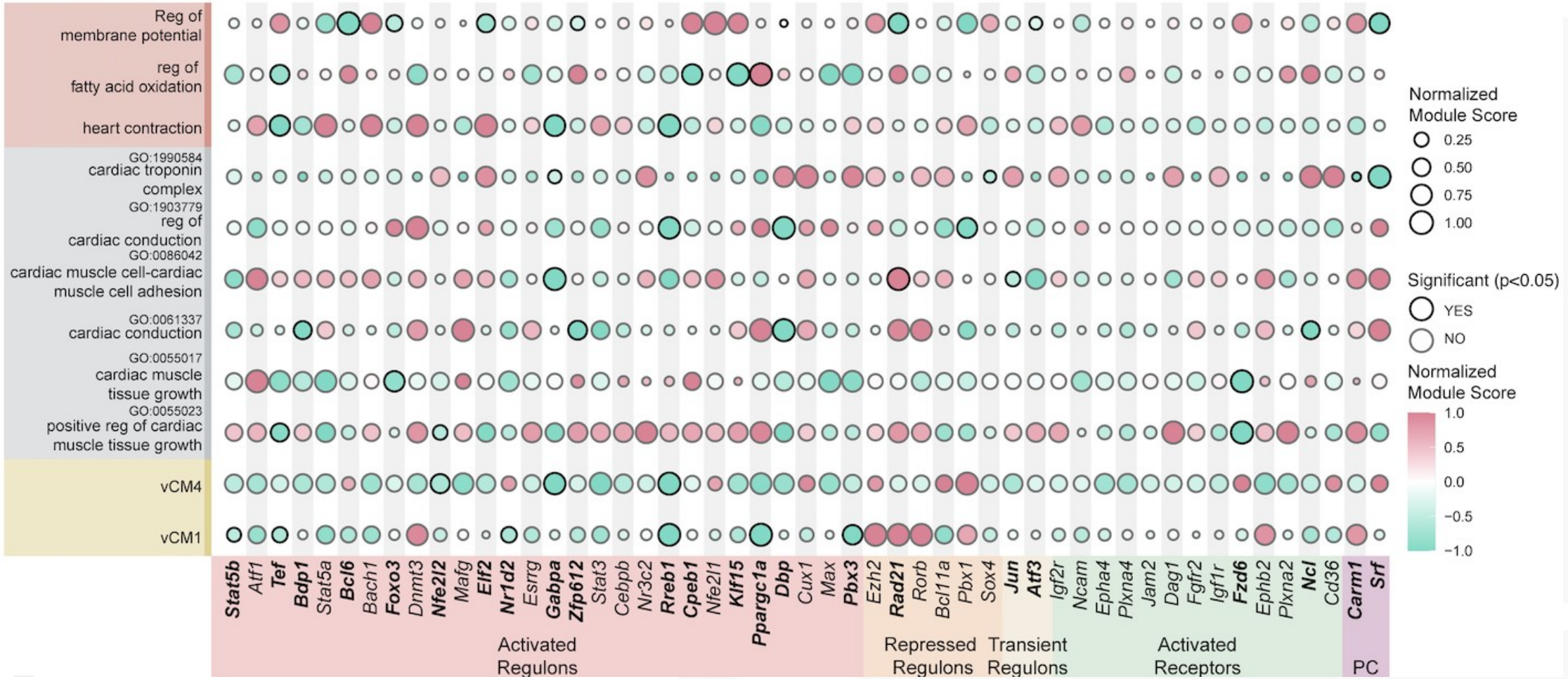




50 candidate regulators; 5,706 guide-assigned nuclei.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 6d; published screen Results

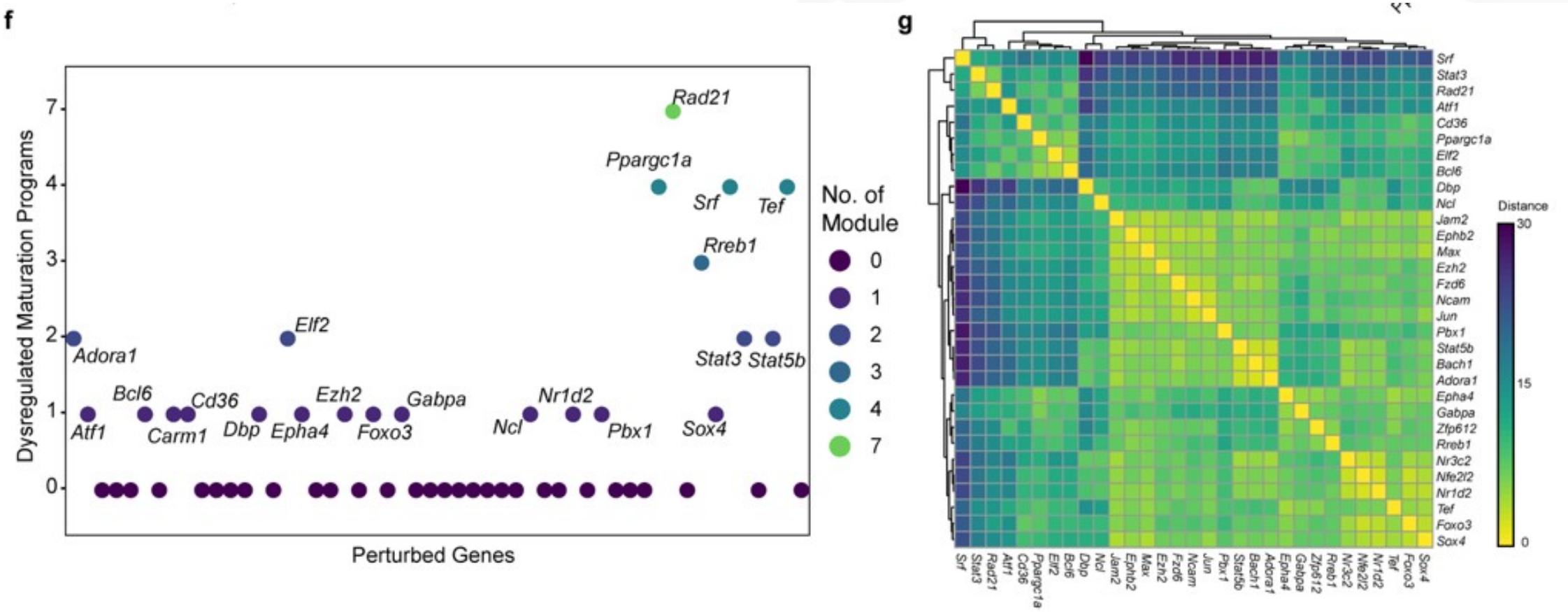
PIP-seq reveals program-specific effects of candidate regulators



Distinct gene perturbations produce distinct maturation-program fingerprints.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 6e; original perturbation matrix

Perturbation fingerprints identify broad and selective regulators

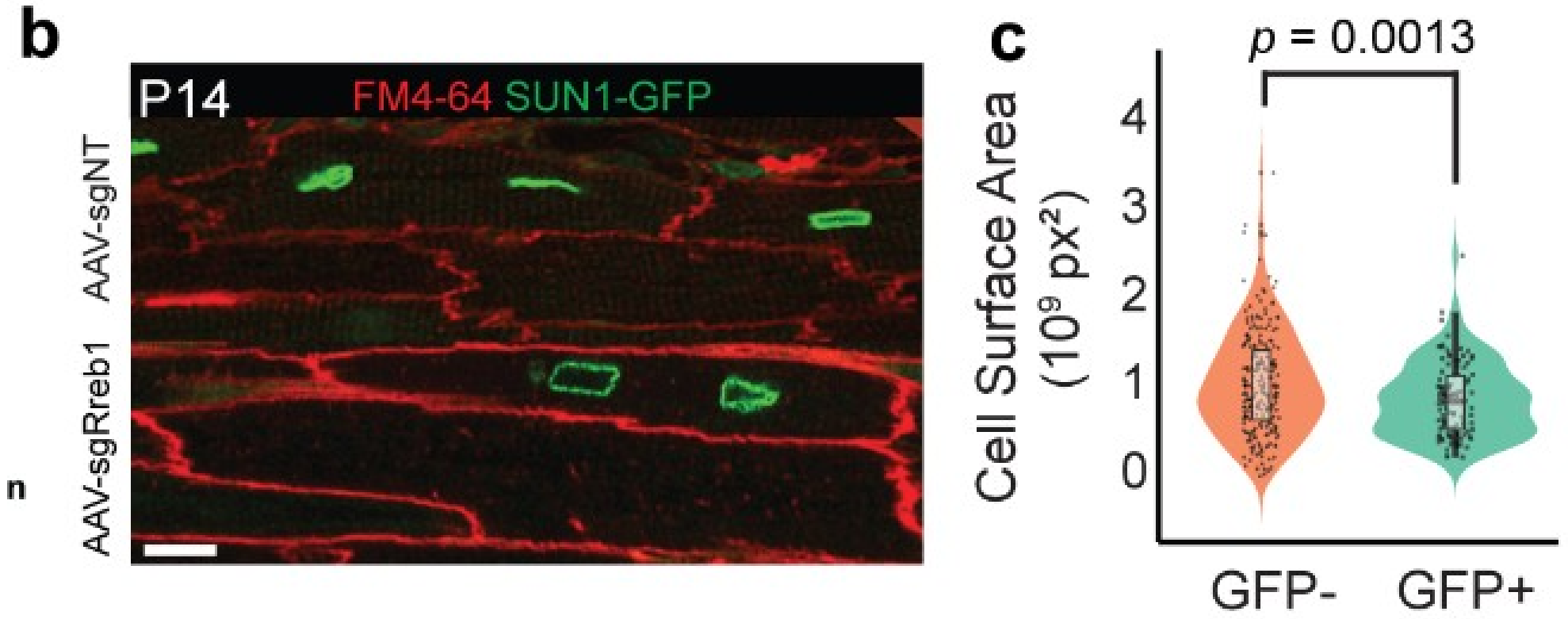


The screen prioritizes genes and the functional measurements needed to test them.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 6f,g

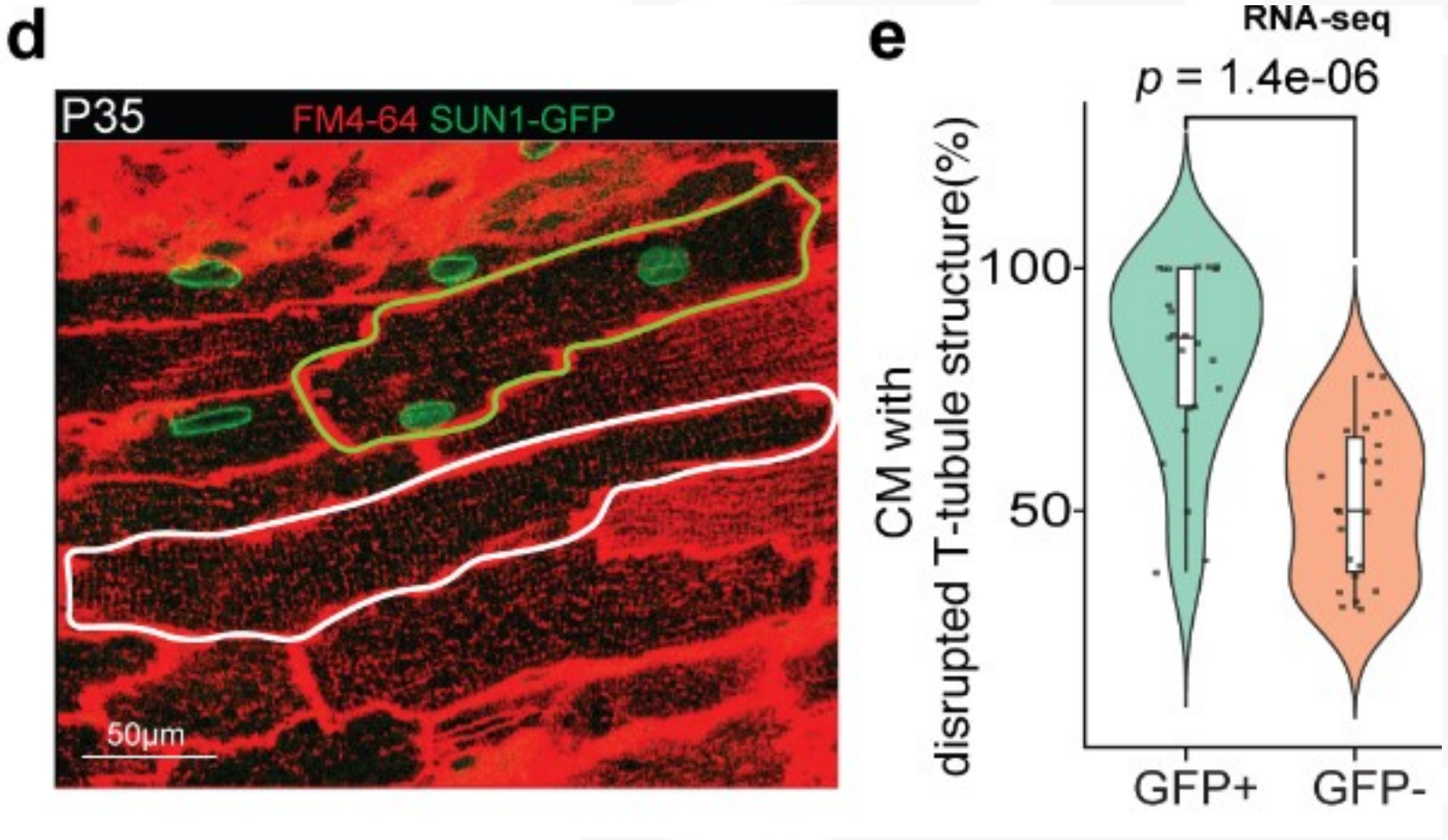
Postnatal Rreb1 perturbation impairs CM growth

P0 · AAV perturbation → **P14 · CM growth** → **P35 · CM / heart function**



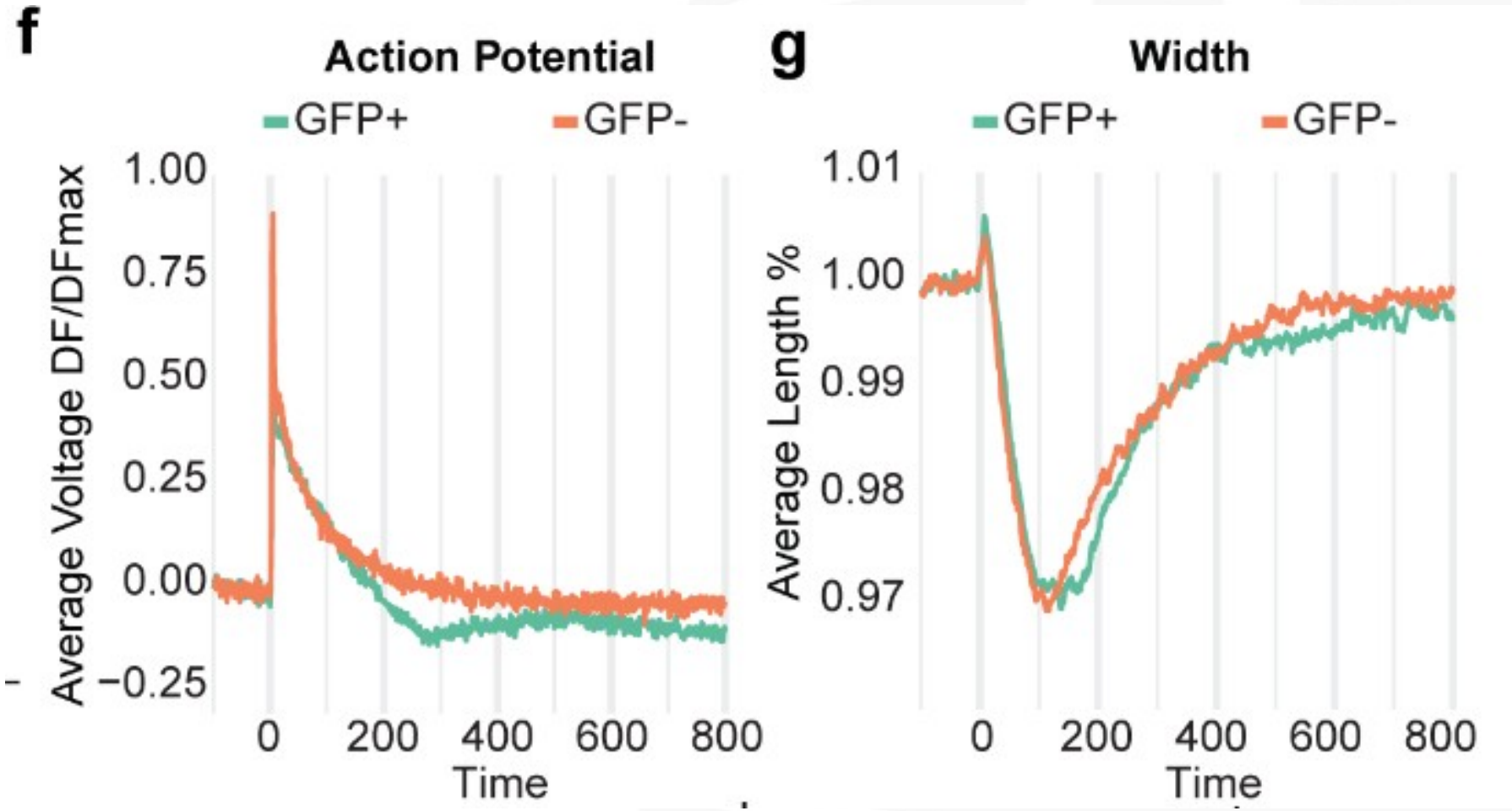
A postnatal intervention isolates a developmental role obscured by embryonic lethality.

Rreb1 is required for normal T-tubule organization



Structural validation extends beyond cell size to T-tubule architecture.

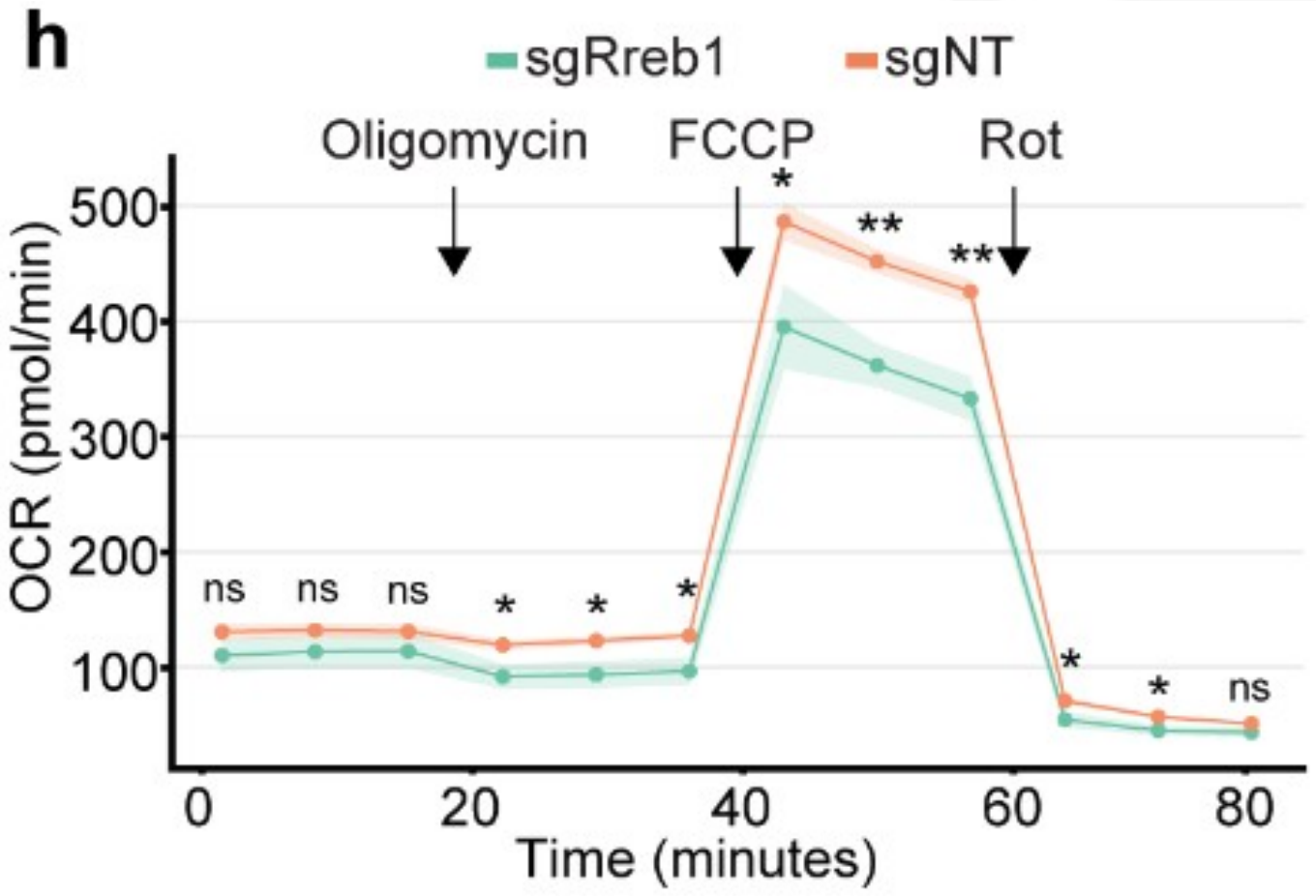
Rreb1 loss alters electrical and contraction readouts



Independent cellular measurements connect the RNA phenotype with CM performance.

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 7f,g

Rreb1 loss reduces mitochondrial respiratory capacity



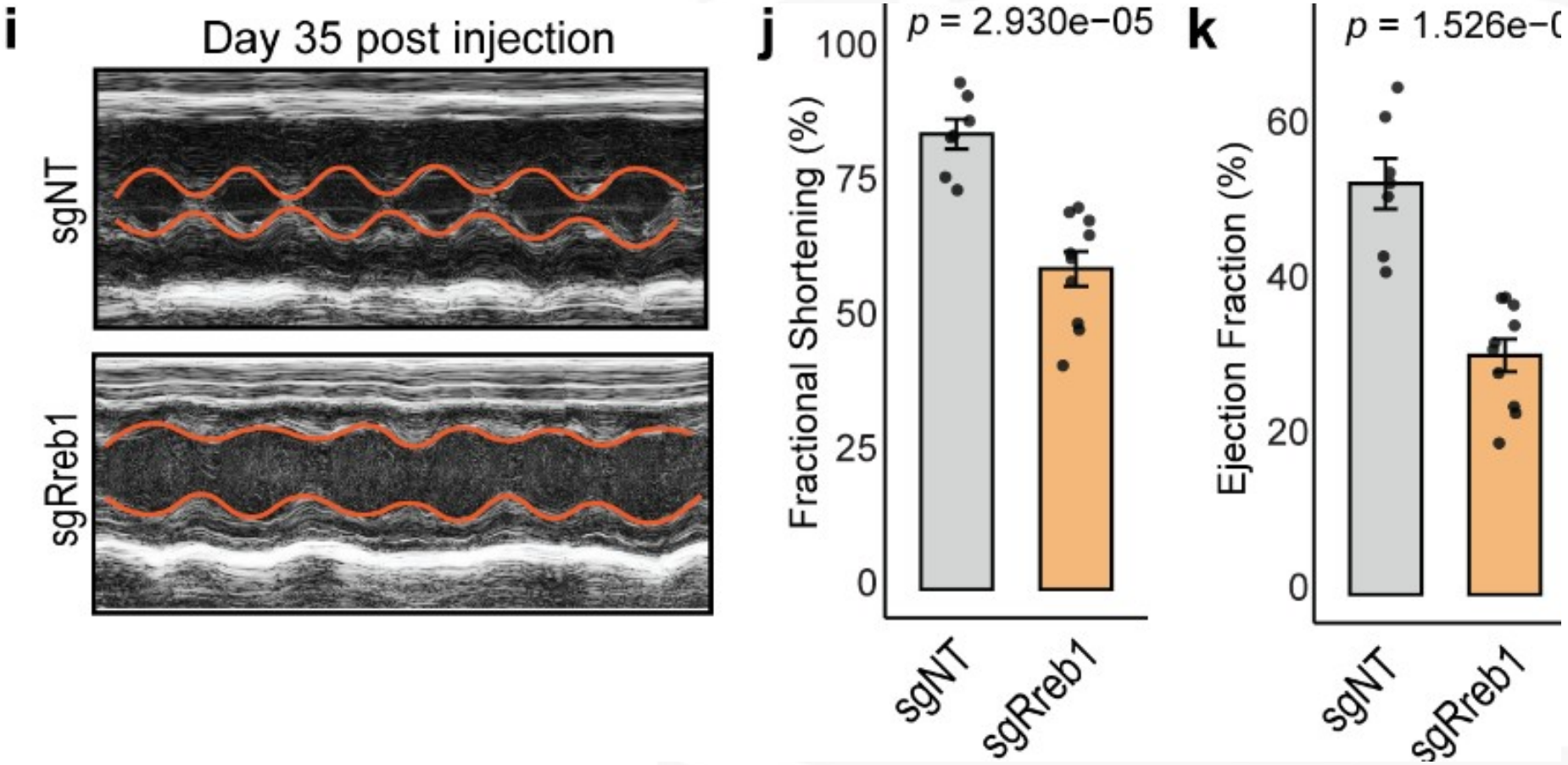
Reduced maximal mitochondrial respiration

A functional metabolic phenotype

Metabolic validation complements structural and electrical measurements.

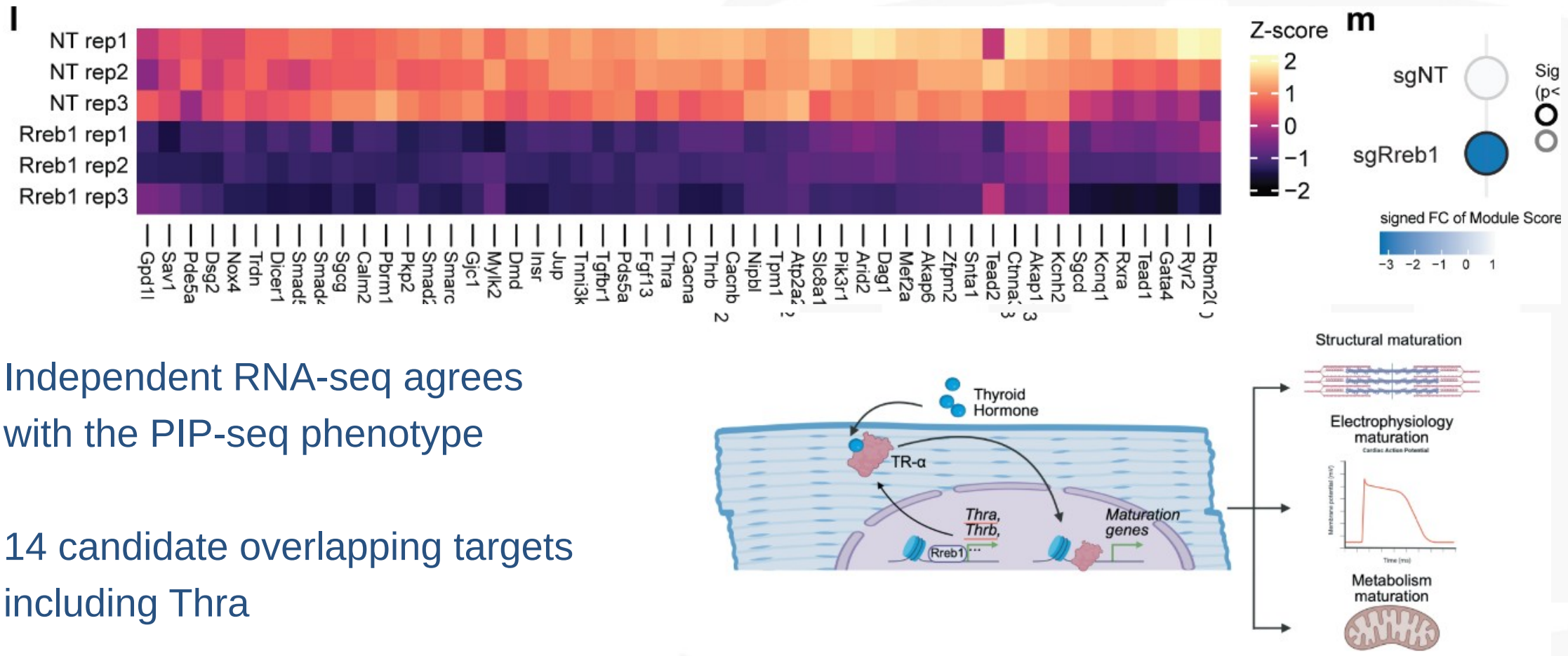
Rreb1 perturbation compromises heart function

P35 echocardiography · sgNT: 7 mice · sgRreb1: 10 mice



Cellular phenotypes are accompanied by reduced pump performance.

Rreb1 supports a maturation-linked transcriptional program



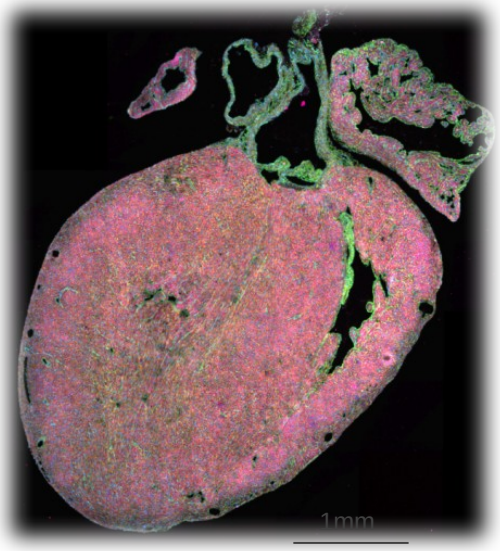
Independent RNA-seq agrees with the PIP-seq phenotype

14 candidate overlapping targets including Thra

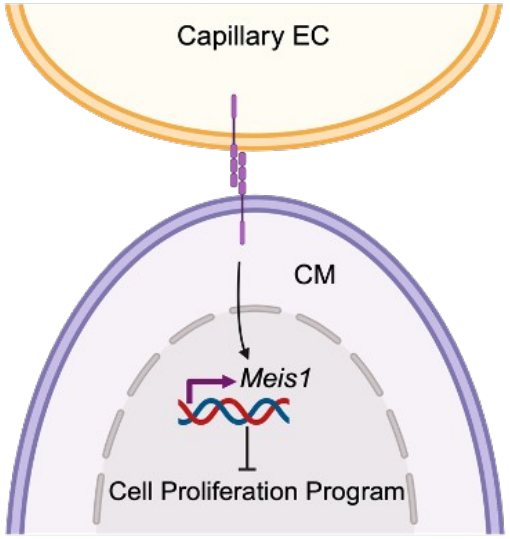
PIP-seq, independent RNA-seq and regulatory inference converge on a testable model.

A spatial atlas and PIP-seq reveal regulators of CM maturation

A single-cell atlas of postnatal hearts with spatial and temporal resolution

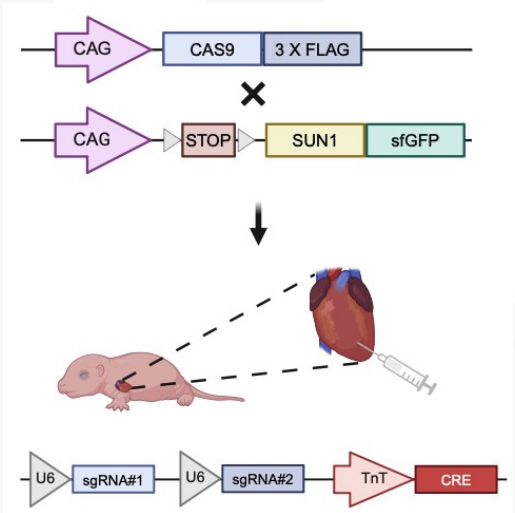


Spatial and temporal atlas of postnatal hearts

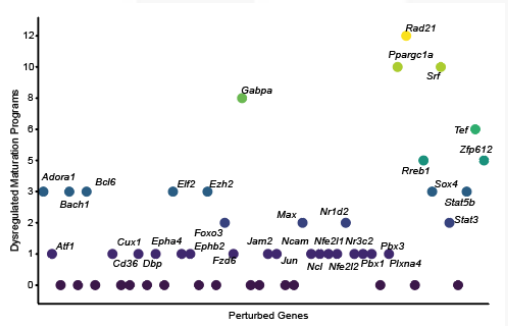


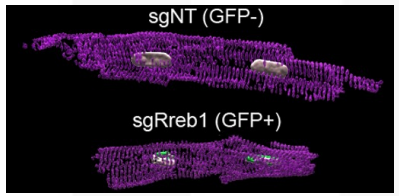
Intercellular communications

in vivo probe-based indel-detectable perturb-seq



in vivo Probe-based Indel-detectable Perturb-seq





Novel regulators for the postnatal heart maturation

Wang H*, Dong Y*, Song Y* et al., NCVR. 2026

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American Heart Association®

Backup: the paper's 21-gene maturation index

Rxrg	Esrrg	Jph2
Ppargc1a	Mitf	Bin1
Ppargc1b	Myh6	Lpin1
Cebpa	Tnni3	Pln
Cebpb	Myl2	Ppara
Esrra	Kcnq1	Ppard
Esrrb	Jph1	Cpt1b

Published Methods: Seurat AddModuleScore

Independent developmental benchmark; mean leave-one-out r = 0.997

Backup: probe readout and targeted validation

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Detectable perturbation-detection probes showing at least 50% reduced signal

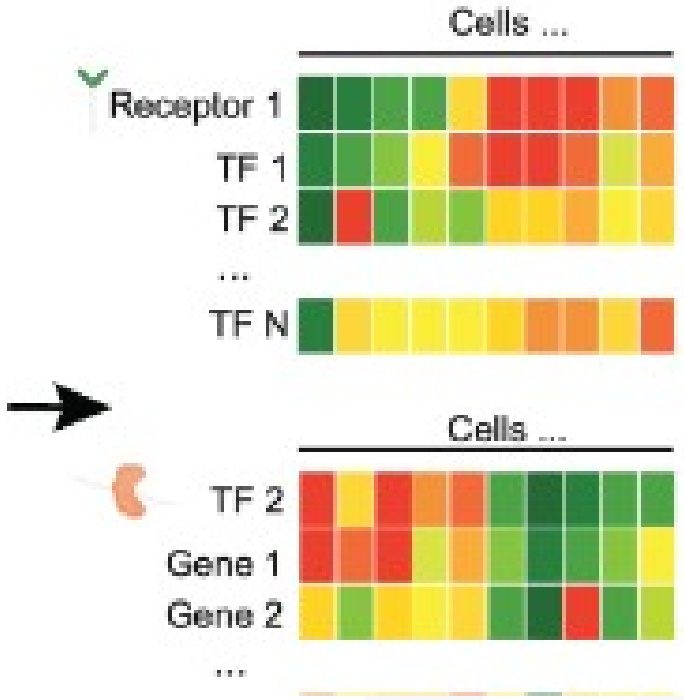
RNA-based perturbation evidence is followed by independent target and functional validation.

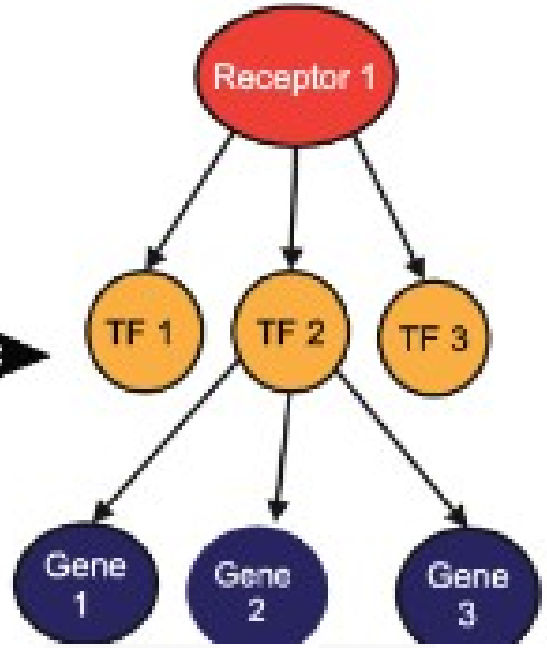
Backup: interpreting the communication and regulatory models

Model distance penalties: paracrine >200 µm; juxtacrine >50 µm

e

	Gene 1	Gene 2	…	Gene 25094
Spatial Cell 1	1	2	…	3
Spatial Cell 2	4	1	…	0
Spatial Cell 3	0	2	…	2
Spatial Cell 4	5	0	…	0
…	…	…	…	…
Spatial Cell 46747	1	3	…	0





Spatial gene expression | **SCENIC regulatory programs** | **Receptor–TF links**

Wang, Dong, Song et al. Nature Cardiovascular Research (2026) | Fig. 2d; Fig. 3e; Methods